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Dear Editors-in-Chief

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Daekook Kang

A strategic analysis of renewable energy sources for hydrogen production based on an augmented fuzzy decision approach

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Abstract

Hydrogen is the cleanest fuel and its generation from renewable sources is essential for combating energy and climate challenges. Meanwhile, the overall utilization of India's renewable energy capacity could facilitate the production of environmentally benign hydrogen in the international market. However, to accelerate the large-scale manufacturing process, it is crucial to examine the potential of renewable energy sources under multiple limits. To do so, this study developed a fuzzy logic-based integrated multistage assessment model to investigate the possibility of renewable energy based H_2 generation in India. Also, the novel notion of bipolar complex linear diophantine fuzzy sets for navigating both favourable and negative views associated to human judgements is addressed. The addition of bipolarity to the complex-valued structure of the linear diophantine fuzzy set improves the information accuracy to be stated precisely. Furthermore, a modified complex bipolar linear diophantine fuzzy sets based Level based weight assessment-Multi-normalization Multi-distance assessment approach is proposed as a multi-criteria decision making tool for evaluating a country's significant renewable energy alternatives based on technical, economic, social, political, and environmental limits, as well as their 14 sub-criteria. The findings demonstrate the validity of the model in identifying an adaptable energy source for H_2 production in the country. In addition, a comparison and sensitivity analysis were carried out to ensure the effectiveness of the strategy.

Keywords: Multi-criteria decision making; fuzzy logic; renewable energy; climate mitigation; energy security; integrated model

1 Introduction

The long-term combustion of conventional fuels for human needs has raised questions regarding environmental concerns. As a result, the need for fuels that are derived from clean and reliable sources is extremely important. Switching to renewable energy sources (RES) for electricity generation delivers cleaner fuel options that are cheaper and more sustainable. To mitigate climatic changes and accelerate the usage of renewable fuels, hydrogen has been regarded as either an alternate fuel for electricity production as it produces water vapour as a by-product or a versatile energy carrier where energy can be stored and delivered when

needed using fuel cell or compression ignition engine[1]. As a result, the demand for hydrogen in chemical industries, oil refineries, and steel production has increased considerably in recent years [2]. The nomenclature used for the study is given in Table 1.

Table 1: Abbreviations

RES	Renewable energy source
RE	Renewable energy
MCDM	Multi-Criteria decision making
GH	Green hydrogen
WtE	Waste to Energy
BFS	Bipolar fuzzy set
CIFS	Complex Intuitionistic fuzzy set
CFS	Complex Fuzzy Sets
BCFS	Bipolar complex fuzzy set
BCIFS	Bipolar complex intuitionistic fuzzy set
LDFS	Linear diophantine fuzzy set
CLDFS	Complex Linear diophantine fuzzy set
BCLDFS	Bipolar Complex Linear diophantine fuzzy set
BCLDFN	Bipolar Complex Linear diophantine fuzzy number
BCLDHFwGAO	Bipolar Complex Linear Diophantine fuzzy weighted geometric aggregation operator
LBWA	Level based weight assessment
TRUST	MuTi-noRmalization mUti-distance aSessmentT

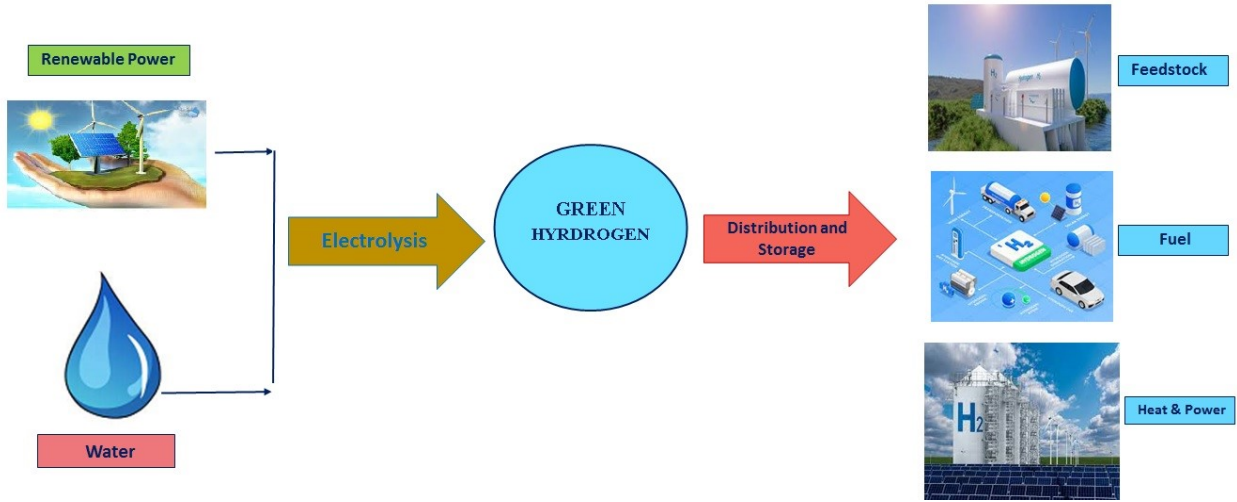


Figure 1: Production and utilization of green hydrogen

Hydrogen can either be produced from hydrocarbons or the renewables. Though hydrogen is regarded as sustainable fuel option, in the present scenario, 47% of the global hydrogen production is from natural gas, 27% from coal, 22% from oil (as a by-product) and only around 4% comes from electrolysis[3] which is again responsible for global emission of

CO₂. However, large-scale deployment of hydrogen generation induced by RES is a green and environmentally benign option that will be essential to achieving net zero emissions [4]. The production and the utilization of renewable hydrogen is depicted in figure 1. Several countries, including Japan, South Korea, the United States, and Australia, have devised plans that include hydrogen as a primary driver for carbon neutrality [5]. These strategies or policies mainly focus on hydrogen as an industrial and automotive fuel [6]. On the other hand China is the leading producer of hydrogen which accounts to 18% in the total production of H₂ in the world [7]. Green hydrogen (GH) is produced through electrolysis of water where the electricity required for the process is delivered from the RE sources [8]. However, the production cost, feedstock availability, storage and transportation of H₂ act as major challenges for the wide spread deployment of H₂ economy [9].

India one of the fast growing economies, is one of the largest importer of fossil fuel with coal and oil satisfying 74% of the total energy demand in the country [10]. As of 2023, the population of India has surpassed that of China[11]. To overcome the energy scarcity faced in the country, the outlook for alternative sources for satisfying the rising energy demand is accelerating [12]. India ranks fourth in the global RE based power generation with 40% of the energy demand is being fulfilled by it. As, the RE based techniques are expected to dominate the future energy mix of India, H₂ will have a major role to play in it. As the role of H₂ in the global energy system is predominant now, India also developed anticipating strategies and polices in line with the transforming energy economy. The government has supported various hydrogen initiatives at universities, scientific institutions, and businesses to investigate the technological and commercial feasibility of integrating hydrogen into the energy grid.

Hydrogen is mostly produced in India in the chemical fertiliser industry, petroleum refineries, and as a byproduct in the chlor-alkali sector [13, 14]. The quantity of hydrogen needed in these sectors is expected to increase to 137 megatons (Mt) in 2050 considering future expansion [15]. Currently, the hydrogen produced comes from natural gas as a primary source with 50% of it being imported for industrial purposes [16]. The industrial and the transportation sector occupies 45% and 25% respectively of the total energy consumption [17]. The transportation sector is the largest emitter of GHG which is 14 % of the energy-related CO₂ emissions. In India, the price of H₂ ranges from INR 350 to 400 per kg (\$4.5 to 5.3 per kg). As the variable energy source are expected to have a predominant role in the energy generation by 2030, if only 2% of the energy produced is used for H₂ generation, the

energy produced will be used to fuel 92000 FC vehicles, which in turn will help in decarbonizing the sector [18]. Furthermore, based on the ammonia needed to produce different fertilisers, the amount of hydrogen necessary in the fertiliser sector is calculated. According to the forecast, even 1% of the ammonia produced by using green H_2 can prevent the import of up to 0.4 MMSCFD of NG [16].

Despite significant expansion in the amount of energy generated and distributed across various sectors, supplying power in rural and urban areas remains a difficult undertaking, with the country facing an energy deficit of 2.1% and approximately 20,000 communities remaining in the dark [19]. Geographical limitations pose challenges for the distribution of energy. Such issues might be resolved by using hydrogen-based RES energy generation [20]. India is a country rich in enormous RE sources. The extensive installation of RE-based power systems will lower the cost of electrolyzers, assisting in the expansion of H_2 production. In order to facilitate the implementation of environmentally friendly hydrogen-based processes, it is crucial to select the most suitable RES for its production. This entails the utilisation of a decision algorithm that assesses multidimensional criteria to discover the preferred alternative.

The following is how the paper is structured: Section 2 gives the study's theoretical foundation. Section 3 describes the new concept of BCLDFS, whereas section 4 describes the mathematical approach. A numerical illustration is discussed in section 5. Sections 6 and 7 contain the validation of the result and the conclusion.

Potential of RE sources for H_2 production in India:

The National GH Mission introduced on 2022, focusses on becoming self-sufficient economy dependent on RE sources integrated with green H_2 , which in turn will raise the country's global economic status. It is estimated to increase the GH production capacity of at least 5 Million Metric Tonne (MMT) per annum with an associated RES capacity addition of about 125 GW in the country simultaneously reducing the import cost on fossil fuels by INR. One lakh crore [21]. Currently, India generates around 6.7 Mt of H_2 per year [22]. The RE potential is roughly 900 GW from diverse sources, with solar energy accounting for approximately 53% of the entire RE segment, followed by wind energy (34%), bio power (biomass and WtE) (9%), and small hydro (4%) [23]. The Government of India has allocated INR 25 crore in the Union Budget 2021-22 for hydrogen energy research and development, with the goal of producing three-fourths of its hydrogen from renewable sources by 2050. As a result, the drop in sustainable energy prices, as well as national policy actions, will have

105 a catalytic role in lowering green H_2 production costs by 20 to 30%, reaching INR 230-240
106 per kg, [24]. This bodes well for GH's future involvement in the country's energy mix. The
107 current role of different RE sources in GH drive are discussed as follows:

108 **Solar energy:** Solar energy system is one of India's most notable RES, with a generation
109 capacity of 70.096 GW as of 2023 [25] and a solar insolation of 4-7kWh per Sq.m per day [26].
110 Among the world's largest solar parks, three are located in India: Rajasthan, Karnataka,
111 and Andhra Pradesh, with capacities of 2255 MW, 2000 MW, and 100 MW, respectively
112 [28]. Solar thermal energy, solar photovoltaic (PV), and photo electrochemical energy are
113 the three basic types of solar energy. The electricity generated by a PV source is used to
114 generate H_2 via electrolysis. As of the end of 2021, India is fourth in the world in terms
115 of solar PV deployment. DST-funded initiatives are being conducted at the University of
116 Petroleum and Energy Studies in Dehradun to produce H_2 using a PV electrolyzer system
117 [29]. The efficiency of solar PV hydrogen production is determined by the efficiency of the
118 electrolyzer employed in the process [30]. It is projected that PV-based H_2 fuel cells may
119 generate 1538-1856 kg/yr of H_2 in north east India [20].

120 **Wind energy:** About 43.77GW of energy is generated from the wind sources in the
121 country making it fourth in the total installed capacity of wind based energy production
122 in line with USA, Germany and China [27]. Karnataka, Tamil Nadu and Gujarat are the
123 major contributors of wind power with Tamil Nadu having the largest wind energy capacity
124 of about 9.23 GW [28]. Wind speed in these areas are more than 5.4 m/s making them
125 economically viable for wind based energy production [31]. It is estimated that a 4GW wind
126 farm can produce enough hydrogen to power 1.4 million homes [32]. The gross wind power
127 capacity installed in the country promise an effective platform for H_2 generation through
128 electrolytic process [21]. The world's first project capable of producing GH directly with the
129 help of onshore wind turbine through electrolysis is under development by Siemens Gamesa
130 in Western Denmark [32]. Several onshore and offshore wind projects for H_2 production are
131 under development in Australia, Germany, Chile, Denmark and China [32].

132 **Biomass:** Around 140 billion tonnes of biomass are produced annually throughout the
133 world with wood waste, agricultural waste and residual waste making up the major compo-
134 sition [33]. India produces about 750 million metric tonnes of biomass annually accounting
135 to 10.248 GW power production [25]. It is estimated that India has a potential to generate
136 28GW of power using 230 million metric tonnes of agro residues and 14GW using bagasse
137 based cogeneration with sugar mills [34]. Maharashtra has the maximum biomass energy ca-

138 capacity of 2632 MW [28]. About 32% of the primary energy sources are derived using biomass
139 and 70% of the population depends on biomass for their energy needs [34]. The mass % of H₂
140 in biomass is 6%, equating to 0.672 m³ gaseous H₂ generated per kg of biomass, accounting
141 for more than 40% of the total energy of biomass [33]. Several biological and thermochemical
142 processes are available for H₂ production from biomass [35]. The plant of biological produc-
143 tion of hydrogen was set up at the Murugappa Chettiar Research Centre (MCRC) Chennai
144 [36]. Several DST funded projects are being carried out in several institutions for producing
145 hydrogen from biomass [36].

146 **Hydro energy:** India holds the fifth largest installed capacity of hydropower plants with
147 46.850GW power production [25]. These plants are divided into small (< 25 MW) and large
148 (> 25MW) hydro plants based on the production capacity [37]. The northern and north
149 eastern states contribute to about 79% of the total hydropower capacity of the country [38].
150 There are around 4,877 constructed and 312 under-constructed dams in India. The run of
151 rivers, dams and pumped hydro storage are the primary sources for hydropower plants. As
152 the sources are not intermittent or not affected by the weather they can be directly used for
153 H₂ production through electrolytic process.

154 **Waste to Energy:** Waste-to-energy (WtE) is the process of converting waste into
155 electricity and heat, or the path of turning waste into a fuel source. India generates 62
156 million tonnes of solid waste with an annual increase of 4%. This means that the market
157 size of WtE sector is projected to be 14 billion USD by 2025 with an annual increase of
158 around 7%. This untapped potential can be used in the production of H₂. Currently, they
159 possess an installed capacity of 565.58 MW power production [25]. These plants focusses
160 on recovering energy in the form of either biogas/syngas, bioCNG or electricity. Biogas is
161 a combination of CH₄ and CO₂, including other gas traces like H₂S, H₂, N₂, NH₃, H₂O and
162 O₂ which can be used for H₂ production through the decomposition of methane [39, 40].

163 2 Background of the study

164 In a decision process, the nature of human thoughts, which are a compilation of positive
165 and negative effects (two sided), and the ambiguous nature of the problems make attaining
166 desirable solutions a difficult task. The unipolar sets, like the fuzzy sets, were able to handle
167 only the fuzziness by assigning grades of membership to each element of the set between [0, 1]
168 [41]. As a result, the concept of bipolarity fused with fuzzy set theory (BFS) was developed,

169 which simultaneously addresses the polarity aspects and the ambiguity of the problem more
170 precisely by assigning values between $[-1, 1]$ [42, 43]. On the other hand Ramot et al. [44]
171 proposed the notion of complex fuzzy sets (CFS) which assigned membership values in the
172 unit circle of the complex plane rather than $[0,1]$. The popularity of this notion lies in
173 the introduced phase term being added to the amplitude or the membership grade which
174 was able to deal with the perplexity in expressing the complex valued functions in human
175 language. As a result, several researchers have worked on this concept and have discussed
176 their properties and applications in various fields [45, 46, 47]. Alkouri et al.[48] developed
177 the complex intuitionistic fuzzy sets (CIFS) by adding non-membership grades to the CFS.
178 Further, the versatility of the developed CIFS in handling vagueness in decision process was
179 examined by Rani et al. [49] and Garg et al.[50].

180 In the meantime, bipolar complex fuzzy sets (BCFS) which expresses the amplitude and
181 the phase terms in the form of BFS was developed as an effective tool for handling vagueness
182 in complex problems [51]. Additionally Gulistan et al. [52] developed the distance measure
183 for BCFS and employed in evaluating the transport problem. Knowing the advantages
184 of the BCFS, several works regarding BCFS have been developed [53, 54]. Moreover Al-
185 Husban et al. [55] introduced the bipolar complex intuitionistic fuzzy sets (BCIFS) which
186 generalizes the properties of both IFS and BCFS and similar studies emerged in the topic
187 [56]. Linear diophantine fuzzy sets (LDFS) were developed as a generalization of the IFS with
188 the introduction of constrained parameters to the existing membership and non-membership
189 grades [57]. This motivated research in this field and several extensions like dual hesitant
190 LDFS [58], spherical LDFS [59], q-LDFS [60], complex LDFS [61], linguistic LDFS [62] were
191 discussed in establishing vagueness in MADM problems.

192 When working with a problem with multi objectives subject to certain constraints that
193 requires two way assessment of the situation (positive and the counter effect) i.e., consider
194 the situation where a manufacturing company wants to decide on a project for investment.
195 The optimal project is chosen based on the associated profit and risk factors simultaneously
196 with the constraints “very beneficial” and “less beneficial”. Here, the decision has to be made
197 assessing positive economic and its negative impact, positive and negative social and political
198 factors. In such situation the use of IFS, CLDFS, BFS, LDFS fails to express the fuzziness
199 in a precise manner. However, there lacks a study articulating the fuzziness (CLDFS) and
200 the polarity (BFS) for efficient analysis of the situation at hand. This necessitates the need
201 for combining BFS with CLDFS which additionally provides positive and negative values to

each of the complex membership, non-membership grades and the constraint parameters in analysing the two sided (positive and negative) information. The novelty of the developed bipolar complex linear diophantine fuzzy sets (BCLDFS) lies in the fact that it is able to deliver the two sided nature of the information combined with the fuzziness of the problem.

Table 2: Fuzzy MADM approaches in RES selection

Authors	Objective of the study	Optimization technique	Country	Result
Goswami et al. [74]	RE based power plant selection	MEREC-PIV tool	India	Hydroelectric power plant is the appropriate option whereas Biomass plant is the last option
Sarkodie et al.[75]	RE based source selection for power generation	CRITIC-MOORA, TOPSIS and COPRAS	Ghana	Hydro power is identified as the most relevant choice
Xu et al. [76]	Identification of ideal RE source for promoting H2 economy	AHP-DEA	Pakistan	Solar and Wind energy are equally reliable options whereas geothermal is the least efficient
Abdel-Basset et al. [77]	Evaluation of RE based sustainable H2 production	Triangular fuzzy AHP-TOPSIS	China	Solar energy is the suitable choice
Messaoudi et al. [78]	Land suitability for solar based H2 production	AHP-GIS	Algeria	The technical and economic constraints has played a crucial role in land estimation
Sarat et al. [79]	Multidimensional analysis of the sustainable energy sources	AHP-TOPSIS	India	Solar energy followed by wind and hydro power
Chen et al. [80]	Multistage analysis of green energy sources	Picture linguistic fuzzy based PROMETHEE-II method	Northwest China	Solar energy is the best choice
Liu et al. [81]	Optimal RES selection for building H2 economy	Fuzzy AHP-TOPSIS	China	Solar energy is the desirable option

Weight determination of the criteria has a significant effect on the outcome of the decision process. Zavidoc et al. [63] suggested the Level Based Weight Assessment (LBWA) model, which assigns weights to the criteria based on experts standpoint. In contrast to other subjective methods like AHP and BWM, where the pairwise comparison of criteria based on ordinal scale becomes complex requiring further modification for retaining the consistency of the result, the developed method groups the criteria based on their significant levels, making it easily accessible for a wide range of criteria [64]. The method has been combined with BWM method for evaluating the health sectors in Eastern Europe [65]. Further Biswas et al. [66] developed the spherical fuzzy based LBWA-MULTIMOOSRAL approach for assessing the leanness of Small, and Medium Enterprises (MSMED) in India. Further the method has been incorporated in gauging MCDM problems like firing position selection [67], oil spill response strategy selection [68] and container port evaluation for shipping companies [69].

Among the available distance based approaches for prioritizing alternatives, TRUST (MulTi-noRmalization mUlti-distance aSsessmentT) is a recently developed MADM tech-

221 nique which uses multi-stage normalization and multi-distance measures for evaluating the
 222 options [70]. The advantage of using multistage processing of information is that it helps
 223 in minimizing the errors and reduces the influence of using a single distance measure. The
 224 method is employed in evaluating the battery swapping stations. To further improve the
 225 method's efficacy Bonab et al. [71] suggested fuzzy based TRUST method for sustainable
 226 supplier selection. In the meantime, the method was fused with fuzzy CRADIS (Compromise
 227 Ranking of Alternatives from Distance to Ideal Solution) for identifying the ideal supplier
 228 in agribusiness [72]. However, the existing TRUST method suffers limitations which hinders
 229 its wide spread application. For instance, the use of logarithmic normalization can only be
 230 used for standardizing positive information limiting its hand in assessing bipolar information.
 231 Also, the distance measures proposed loses their versatility when dealing with CBF informa-
 232 tion. In order to get over this drawback and increase the approach's adaptability, this study
 233 creates an enhanced TRUST method based on BCLDFS and investigates its effectiveness
 234 when confronting volatile settings.

235 Considering MCDM as prominent methodology for handling multifaceted problem, stud-
 236 ies have been carried out for analysing the potential of RE strategies for electricity and
 237 hydrogen generation [64, 73]. Some of these studies are presented in Table 2.

238 Identifying potential RES for GH production is crucial for advancing the country's GH
 239 agenda. Significant investments and research breakthroughs in the hydrogen economy and
 240 green energy sources emphasise the need for a multifaceted appraisal of the situation. As
 241 multiple research offer various energy sources for H_2 production, an analysis of the problem
 242 in the Indian context is required. The study proposes a BCLDF-based LBWA and improved
 243 fuzzy TRUST approach for analysing the five RES under five criterion and fourteen sub-
 244 criteria.

245 **2.1 Motivation for the study**

- 246 • The two way assessment (i.e., the positive and counter effect) of the situation is very
 247 effective when dealing with complex situation. Even so, there is a dearth of research
 248 combining complexity and polarity into a single fuzzy structure, which requires further
 249 investigation.
- 250 • The inclusion of control parameters into the LDFS has extended its role in expressing
 251 ambiguous information to an extent. The advantages of BFS highlight further insight

into the idea of fusing it to the CLDFS, which would benefit the advantages of both sets.

- Fuzzy LBWA for weight assessment combined with TRUST, a MCDM model in the context of CLDFS setting has not been discussed so far
- Owing to the advantages of Hydrogen produced through renewables, from the table 2, there lacks study incorporating a fuzzy decision model for analysing the RE potential under specific constraints for the production of hydrogen in India.

2.2 Research Objectives

- This work introduces the innovative notion of BCLDFS, a hybrid of BFS and CLDFS, as an outstanding tool for dealing with complicated and disparate information.
- To circumvent the confines of the earlier TRUST technique, the study developed an enhanced paradigm based on the BCLDF logic as a malleable decision tool for prioritizing potential options.
- The usefulness of the suggested fuzzy LBWA-TRUST model in the BCLDF context in analysing the green energy source for producing hydrogen is demonstrated.

3 Preliminaries

Definition 3.1. Let Δ be a reference set. A bipolar complex linear diophantine fuzzy set (BCLDFS) $\aleph$ on Δ is characterized by

$$\aleph = \{(\tau_b, \langle \Gamma_{\aleph}^+(\tau_b)e^{i2\pi(\zeta_{\Gamma_{\aleph}}^+(\tau_b))}, \Gamma_{\aleph}^-(\tau)e^{i2\pi(\zeta_{\Gamma_{\aleph}}^-(\tau_b))}, \Lambda_{\aleph}^+(\tau_b)e^{i2\pi(\zeta_{\Lambda_{\aleph}}^+(\tau_b))}, \Lambda_{\aleph}^-(\tau)e^{i2\pi(\zeta_{\Lambda_{\aleph}}^-(\tau_b))} \rangle, \langle \Phi_{\aleph}^{b+}e^{i2\pi(\zeta_{\Phi_{\aleph}}^+(\tau_b))}, \Phi_{\aleph}^{b-}e^{i2\pi(\zeta_{\Phi_{\aleph}}^-(\tau_b))}, \Psi_{\aleph}^{b+}e^{i2\pi(\zeta_{\Psi_{\aleph}}^+(\tau_b))}, \Psi_{\aleph}^{b-}e^{i2\pi(\zeta_{\Psi_{\aleph}}^-(\tau_b))} \rangle) : \tau_b \in \Delta\} \quad (1)$$

Here, $\Gamma_{\aleph}^+(\tau_b)e^{i2\pi(\zeta_{\Gamma_{\aleph}}^+(\tau_b))}$, $\Lambda_{\aleph}^+(\tau_b)e^{i2\pi(\zeta_{\Lambda_{\aleph}}^+(\tau_b))}$, $\Phi_{\aleph}^{b+}e^{i2\pi(\zeta_{\Phi_{\aleph}}^+(\tau_b))}$, $\Psi_{\aleph}^{b+}e^{i2\pi(\zeta_{\Psi_{\aleph}}^+(\tau_b))}$, represent the positive grades of complex valued satisfaction, dissatisfaction and the control parameters of $\tau_b \in \Delta$ associated with $\aleph$ and $\Gamma_{\aleph}^-(\tau)e^{i2\pi(\zeta_{\Gamma_{\aleph}}^-(\tau_b))}$, $\Lambda_{\aleph}^-(\tau)e^{i2\pi(\zeta_{\Lambda_{\aleph}}^-(\tau_b))}$, $\Phi_{\aleph}^{b-}e^{i2\pi(\zeta_{\Phi_{\aleph}}^-(\tau_b))}$, $\Psi_{\aleph}^{b-}e^{i2\pi(\zeta_{\Psi_{\aleph}}^-(\tau_b))}$ are the negative grades of complex valued satisfaction, dissatisfaction and the control parameters of $\tau_b \in \Delta$ associated with $\aleph$ subject to the constraints:

$$(i) \ 0 \leq \Phi_{\mathfrak{N}}^{b+} + \Psi_{\mathfrak{N}}^{b+} \leq 1$$

$$(v) \ -1 \leq \Phi_{\mathfrak{N}}^{b-} + \Psi_{\mathfrak{N}}^{b-} \leq 0$$

$$(ii) \ 0 \leq \Phi_{\mathfrak{N}}^{b+} \Gamma_{\mathfrak{N}}^+(\tau_b) + \Psi_{\mathfrak{N}}^{b+} \Lambda_{\mathfrak{N}}^+(\tau_b) \leq 1$$

$$(vi) \ 0 \leq \Phi_{\mathfrak{N}}^{b-} \Gamma_{\mathfrak{N}}^-(\tau_b) + \Psi_{\mathfrak{N}}^{b-} \Lambda_{\mathfrak{N}}^-(\tau_b) \leq 1$$

$$(iii) \ 0 \leq \zeta_{\Phi_{\mathfrak{N}}^b}^+ + \zeta_{\Psi_{\mathfrak{N}}^b}^+ \leq 1$$

$$(vii) \ -1 \leq \zeta_{\Phi_{\mathfrak{N}}^b}^- + \zeta_{\Psi_{\mathfrak{N}}^b}^- \leq 0$$

$$(iv) \ 0 \leq \zeta_{\Phi_{\mathfrak{N}}^b}^+ \zeta_{\Gamma_{\mathfrak{N}}^+}^+(\tau_b) + \zeta_{\Psi_{\mathfrak{N}}^b}^+ \zeta_{\Lambda_{\mathfrak{N}}^+}^+(\tau_b) \leq 1$$

$$(viii) \ 0 \leq \zeta_{\Phi_{\mathfrak{N}}^b}^- \zeta_{\Gamma_{\mathfrak{N}}^-}^-(\tau_b) + \zeta_{\Psi_{\mathfrak{N}}^b}^- \zeta_{\Lambda_{\mathfrak{N}}^-}^-(\tau_b) \leq 1$$

Furthermore, any element $\mathfrak{d}$ of $\mathfrak{N}$ taking the form $\mathfrak{d} = (\langle (\Gamma_{\mathfrak{N}}^+, \zeta_{\Gamma_{\mathfrak{N}}^+}^+), (\Gamma_{\mathfrak{N}}^-, \zeta_{\Gamma_{\mathfrak{N}}^-}^-), (\Lambda_{\mathfrak{N}}^+, \zeta_{\Lambda_{\mathfrak{N}}^+}^+), (\Lambda_{\mathfrak{N}}^-, \zeta_{\Lambda_{\mathfrak{N}}^-}^-) \rangle, \langle (\Phi_{\mathfrak{N}}^+, \zeta_{\Phi_{\mathfrak{N}}^+}^+), (\Phi_{\mathfrak{N}}^-, \zeta_{\Phi_{\mathfrak{N}}^-}^-), (\Psi_{\mathfrak{N}}^+, \zeta_{\Psi_{\mathfrak{N}}^+}^+), (\Psi_{\mathfrak{N}}^-, \zeta_{\Psi_{\mathfrak{N}}^-}^-) \rangle)$ represent the bipolar complex linear diophantine fuzzy number (BCLDFN). Also, the collection of all BCLDFS over Δ is denoted as $BCLDFS(\Delta)$.

Example 3.1. Let $\Delta = \{\tau_1, \tau_2, \tau_3\}$. Then a BCLDFS $\mathfrak{C}$ on Δ takes the form

$$\mathfrak{C} = \left\{ \left(\tau_1, \langle 0.4e^{i2\pi(0.7)}, -0.6e^{i2\pi(-0.5)}, 0.5e^{i2\pi(0.8)}, -0.7e^{i2\pi(-0.4)} \rangle, \langle 0.3e^{i2\pi(0.8)}, -0.6e^{i2\pi(-0.4)}, 0.6e^{i2\pi(0.2)}, -0.2e^{i2\pi(-0.5)} \rangle \right), \left(\tau_2, \langle 0.5e^{i2\pi(0.4)}, -0.9e^{i2\pi(-0.1)}, 0.7e^{i2\pi(0.7)}, -0.8e^{i2\pi(-0.6)} \rangle, \langle 0.3e^{i2\pi(0.8)}, -0.2e^{i2\pi(-0.5)}, 0.7e^{i2\pi(0.2)}, -0.7e^{i2\pi(-0.1)} \rangle \right), \left(\tau_3, \langle 0.6e^{i2\pi(0.9)}, -0.9e^{i2\pi(-0.6)}, 0.5e^{i2\pi(0.7)}, -0.3e^{i2\pi(-0.4)} \rangle, \langle 0.7e^{i2\pi(0.5)}, -0.4e^{i2\pi(-0.3)}, 0.2e^{i2\pi(0.4)}, -0.4e^{i2\pi(-0.2)} \rangle \right) \right\}$$

Proposition 3.1. Every BCIFN, BCPFN, and BCq-ROFN can be viewed as a BCLDFN, whereas the converse may not be true.

Proof. The proof is similar to that in [61]. □

Definition 3.2. A BCLDFS in the reference set Δ is referred as **absolute BCLDFS**, if it takes the form, $\mathfrak{N}^1 = \{(\tau_b, \langle 1e^{i2\pi(1)}, -1e^{i2\pi(-1)}, 0e^{i2\pi(0)}, 0e^{i2\pi(0)} \rangle, \langle 1e^{i2\pi(1)}, -1e^{i2\pi(-1)}, 0e^{i2\pi(0)}, 0e^{i2\pi(0)} \rangle) : \tau_b \in \Delta\} = \{(\tau_b, \langle 1, -1, 0, 0 \rangle, \langle 1, -1, 0, 0 \rangle : \tau_b \in \Delta)\}$ and is said to be **null or empty BCLDFS** if $\mathfrak{N}^0 = \{(\tau_b, \langle 0e^{i2\pi(0)}, 0e^{i2\pi(0)}, 1e^{i2\pi(1)}, -1e^{i2\pi(-1)} \rangle, \langle 0e^{i2\pi(0)}, 0e^{i2\pi(0)}, 1e^{i2\pi(1)}, -1e^{i2\pi(-1)} \rangle) : \tau_b \in \Delta\} = \{(\tau_b, \langle 0, 0, 1, -1 \rangle, \langle 0, 0, 1, -1 \rangle : \tau_b \in \Delta)\}$.

3.1 Some basic operations on BCLDFS

Definition 3.3. Let $\mathfrak{N}_j \in CLDFS(\Delta)$ of the form

$$\mathfrak{N}_j = \{(\tau_b, \langle \Gamma_{\mathfrak{N}_j}^+(\tau_b)e^{i2\pi(\zeta_{\Gamma_{\mathfrak{N}_j}^+}^+(\tau_b))}, \Gamma_{\mathfrak{N}_j}^-(\tau_b)e^{i2\pi(\zeta_{\Gamma_{\mathfrak{N}_j}^-}^-(\tau_b))}, \Lambda_{\mathfrak{N}_j}^+(\tau_b)e^{i2\pi(\zeta_{\Lambda_{\mathfrak{N}_j}^+}^+(\tau_b))}, \Lambda_{\mathfrak{N}_j}^-(\tau_b)e^{i2\pi(\zeta_{\Lambda_{\mathfrak{N}_j}^-}^-(\tau_b))} \rangle, \langle \Phi_{\mathfrak{N}_j}^{b+}e^{i2\pi(\zeta_{\Phi_{\mathfrak{N}_j}^b}^+)} , \Phi_{\mathfrak{N}_j}^{b-}e^{i2\pi(\zeta_{\Phi_{\mathfrak{N}_j}^b}^-)} , \Psi_{\mathfrak{N}_j}^{b+}e^{i2\pi(\zeta_{\Psi_{\mathfrak{N}_j}^b}^+)} , \Psi_{\mathfrak{N}_j}^{b-}e^{i2\pi(\zeta_{\Psi_{\mathfrak{N}_j}^b}^-)} \rangle) : \tau_b \in \Delta\}$$

for $j = 1, 2, \dots, n$. Then

- $\aleph_1 \subseteq \aleph_2$ if

$$\left\{ \begin{array}{l} \Gamma_{\aleph_1}^+(\tau_b) \leq \Gamma_{\aleph_2}^+(\tau_b), \zeta_{\Gamma_{\aleph_1}}^+(\tau_b) \leq \zeta_{\Gamma_{\aleph_2}}^+(\tau_b), \Gamma_{\aleph_1}^-(\tau_b) \geq \Gamma_{\aleph_2}^-(\tau_b), \zeta_{\Gamma_{\aleph_1}}^-(\tau_b) \geq \zeta_{\Gamma_{\aleph_2}}^-(\tau_b), \\ \Lambda_{\aleph_1}^+(\tau_b) \geq \Lambda_{\aleph_2}^+(\tau_b), \zeta_{\Lambda_{\aleph_1}}^+(\tau_b) \geq \zeta_{\Lambda_{\aleph_2}}^+(\tau_b), \Lambda_{\aleph_1}^-(\tau_b) \leq \Lambda_{\aleph_2}^-(\tau_b), \zeta_{\Lambda_{\aleph_1}}^-(\tau_b) \leq \zeta_{\Lambda_{\aleph_2}}^-(\tau_b), \\ \Phi_{\aleph_1}^+(\tau_b) \leq \Phi_{\aleph_2}^+(\tau_b), \zeta_{\Phi_{\aleph_1}}^+(\tau_b) \leq \zeta_{\Phi_{\aleph_2}}^+(\tau_b), \Phi_{\aleph_1}^-(\tau_b) \geq \Phi_{\aleph_2}^-(\tau_b), \zeta_{\Phi_{\aleph_1}}^-(\tau_b) \geq \zeta_{\Phi_{\aleph_2}}^-(\tau_b), \\ \Psi_{\aleph_1}^+(\tau_b) \geq \Psi_{\aleph_2}^+(\tau_b), \zeta_{\Psi_{\aleph_1}}^+(\tau_b) \geq \zeta_{\Psi_{\aleph_2}}^+(\tau_b), \Psi_{\aleph_1}^-(\tau_b) \leq \Psi_{\aleph_2}^-(\tau_b), \zeta_{\Psi_{\aleph_1}}^-(\tau_b) \leq \zeta_{\Psi_{\aleph_2}}^-(\tau_b) \end{array} \right\}_{(b \in \mathbb{N})}$$

- $\aleph_1 = \aleph_2$ if

$$\left\{ \begin{array}{l} \Gamma_{\aleph_1}^+(\tau_b) = \Gamma_{\aleph_2}^+(\tau_b), \zeta_{\Gamma_{\aleph_1}}^+(\tau_b) = \zeta_{\Gamma_{\aleph_2}}^+(\tau_b), \Gamma_{\aleph_1}^-(\tau_b) = \Gamma_{\aleph_2}^-(\tau_b), \zeta_{\Gamma_{\aleph_1}}^-(\tau_b) = \zeta_{\Gamma_{\aleph_2}}^-(\tau_b), \\ \Lambda_{\aleph_1}^+(\tau_b) = \Lambda_{\aleph_2}^+(\tau_b), \zeta_{\Lambda_{\aleph_1}}^+(\tau_b) = \zeta_{\Lambda_{\aleph_2}}^+(\tau_b), \Lambda_{\aleph_1}^-(\tau_b) = \Lambda_{\aleph_2}^-(\tau_b), \zeta_{\Lambda_{\aleph_1}}^-(\tau_b) = \zeta_{\Lambda_{\aleph_2}}^-(\tau_b), \\ \Phi_{\aleph_1}^+(\tau_b) = \Phi_{\aleph_2}^+(\tau_b), \zeta_{\Phi_{\aleph_1}}^+(\tau_b) = \zeta_{\Phi_{\aleph_2}}^+(\tau_b), \Phi_{\aleph_1}^-(\tau_b) = \Phi_{\aleph_2}^-(\tau_b), \zeta_{\Phi_{\aleph_1}}^-(\tau_b) = \zeta_{\Phi_{\aleph_2}}^-(\tau_b), \\ \Psi_{\aleph_1}^+(\tau_b) = \Psi_{\aleph_2}^+(\tau_b), \zeta_{\Psi_{\aleph_1}}^+(\tau_b) = \zeta_{\Psi_{\aleph_2}}^+(\tau_b), \Psi_{\aleph_1}^-(\tau_b) = \Psi_{\aleph_2}^-(\tau_b), \zeta_{\Psi_{\aleph_1}}^-(\tau_b) = \zeta_{\Psi_{\aleph_2}}^-(\tau_b) \end{array} \right\}_{(b \in \mathbb{N})}$$

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- Complement of BCLDFS

$$\aleph_j^c = \{(\tau_b, \langle \Lambda_{\aleph_j}^+(\tau_b) e^{i2\pi(\zeta_{\Lambda_{\aleph_j}}^+(\tau_b))}, \Lambda_{\aleph_j}^-(\tau_b) e^{i2\pi(\zeta_{\Lambda_{\aleph_j}}^-(\tau_b))}, \Gamma_{\aleph_j}^+(\tau_b) e^{i2\pi(\zeta_{\Gamma_{\aleph_j}}^+(\tau_b))}, \Gamma_{\aleph_j}^-(\tau_b) e^{i2\pi(\zeta_{\Gamma_{\aleph_j}}^-(\tau_b))}, \\ \langle \Psi_{\aleph_j}^b + e^{i2\pi(\zeta_{\Psi_{\aleph_j}}^b)}, \Psi_{\aleph_j}^b - e^{i2\pi(\zeta_{\Psi_{\aleph_j}}^b)}, \Phi_{\aleph_j}^b + e^{i2\pi(\zeta_{\Phi_{\aleph_j}}^b)}, \Phi_{\aleph_j}^b - e^{i2\pi(\zeta_{\Phi_{\aleph_j}}^b)} \rangle) : \tau_b \in \Delta\}$$

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- Union of BCLDFS

$$\cup_{j=1}^n \aleph_j = \{(\tau_b, \langle \Gamma_{\cup \aleph_j}^+(\tau_b) e^{i2\pi(\zeta_{\Gamma_{\cup \aleph_j}}^+(\tau_b))}, \Gamma_{\cup \aleph_j}^-(\tau_b) e^{i2\pi(\zeta_{\Gamma_{\cup \aleph_j}}^-(\tau_b))}, \Lambda_{\cup \aleph_j}^+(\tau_b) e^{i2\pi(\zeta_{\Lambda_{\cup \aleph_j}}^+(\tau_b))}, \\ \Lambda_{\cup \aleph_j}^-(\tau_b) e^{i2\pi(\zeta_{\Lambda_{\cup \aleph_j}}^-(\tau_b))}, \langle \Phi_{\cup \aleph_j}^b + e^{i2\pi(\zeta_{\Phi_{\cup \aleph_j}}^b)}, \Phi_{\cup \aleph_j}^b - e^{i2\pi(\zeta_{\Phi_{\cup \aleph_j}}^b)}, \Psi_{\cup \aleph_j}^b + e^{i2\pi(\zeta_{\Psi_{\cup \aleph_j}}^b)}, \\ \Psi_{\cup \aleph_j}^b - e^{i2\pi(\zeta_{\Psi_{\cup \aleph_j}}^b)} \rangle) : \tau_b \in \Delta\}$$

where

$$\left\{ \begin{array}{l} \Gamma_{\cup \aleph_j}^+(\tau_b) = \sup_j \Gamma_{\aleph_j}^+(\tau_b), \zeta_{\Gamma_{\cup \aleph_j}}^+(\tau_b) = \sup_j \zeta_{\Gamma_{\aleph_j}}^+(\tau_b), \Gamma_{\cup \aleph_j}^-(\tau_b) = \inf_j \Gamma_{\aleph_j}^-(\tau_b), \zeta_{\Gamma_{\cup \aleph_j}}^-(\tau_b) = \inf_j \zeta_{\Gamma_{\aleph_j}}^-(\tau_b), \\ \Lambda_{\cup \aleph_j}^+(\tau_b) = \inf_j \Lambda_{\aleph_j}^+(\tau_b), \zeta_{\Lambda_{\cup \aleph_j}}^+(\tau_b) = \inf_j \zeta_{\Lambda_{\aleph_j}}^+(\tau_b), \Lambda_{\cup \aleph_j}^-(\tau_b) = \sup_j \Lambda_{\aleph_j}^-(\tau_b), \zeta_{\Lambda_{\cup \aleph_j}}^-(\tau_b) = \sup_j \zeta_{\Lambda_{\aleph_j}}^-(\tau_b), \\ \Phi_{\cup \aleph_j}^+(\tau_b) = \sup_j \Phi_{\aleph_j}^+(\tau_b), \zeta_{\Phi_{\cup \aleph_j}}^+(\tau_b) = \sup_j \zeta_{\Phi_{\aleph_j}}^+(\tau_b), \Phi_{\cup \aleph_j}^-(\tau_b) = \inf_j \Phi_{\aleph_j}^-(\tau_b), \zeta_{\Phi_{\cup \aleph_j}}^-(\tau_b) = \inf_j \zeta_{\Phi_{\aleph_j}}^-(\tau_b), \\ \Psi_{\cup \aleph_j}^+(\tau_b) = \inf_j \Psi_{\aleph_j}^+(\tau_b), \zeta_{\Psi_{\cup \aleph_j}}^+(\tau_b) = \inf_j \zeta_{\Psi_{\aleph_j}}^+(\tau_b), \Psi_{\cup \aleph_j}^-(\tau_b) = \sup_j \Psi_{\aleph_j}^-(\tau_b), \zeta_{\Psi_{\cup \aleph_j}}^-(\tau_b) = \sup_j \zeta_{\Psi_{\aleph_j}}^-(\tau_b) \end{array} \right\}_{(b \in \mathbb{N})}$$

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- Intersection of BCLDFS

$$\cap_{j=1}^n \aleph_j = \{(\tau_b, \langle \Gamma_{\cap \aleph_j}^+(\tau_b) e^{i2\pi(\zeta_{\Gamma_{\cap \aleph_j}}^+(\tau_b))}, \Gamma_{\cap \aleph_j}^-(\tau_b) e^{i2\pi(\zeta_{\Gamma_{\cap \aleph_j}}^-(\tau_b))}, \Lambda_{\cap \aleph_j}^+(\tau_b) e^{i2\pi(\zeta_{\Lambda_{\cap \aleph_j}}^+(\tau_b))}, \\ \Lambda_{\cap \aleph_j}^-(\tau_b) e^{i2\pi(\zeta_{\Lambda_{\cap \aleph_j}}^-(\tau_b))}, \langle \Phi_{\cap \aleph_j}^b + e^{i2\pi(\zeta_{\Phi_{\cap \aleph_j}}^b)}, \Phi_{\cap \aleph_j}^b - e^{i2\pi(\zeta_{\Phi_{\cap \aleph_j}}^b)}, \Psi_{\cap \aleph_j}^b + e^{i2\pi(\zeta_{\Psi_{\cap \aleph_j}}^b)}, \\ \Psi_{\cap \aleph_j}^b - e^{i2\pi(\zeta_{\Psi_{\cap \aleph_j}}^b)} \rangle) : \tau_b \in \Delta\}$$

where

$$\left\{ \begin{array}{l} \Gamma_{\aleph_j}^+(\tau_b) = \inf_j \Gamma_{\aleph_j}^+(\tau_b), \zeta_{\Gamma_{\aleph_j}^+}^+(\tau_b) = \inf_j \zeta_{\Gamma_{\aleph_j}^+}^+(\tau_b), \Gamma_{\aleph_j}^-(\tau_b) = \sup_j \Gamma_{\aleph_j}^-(\tau_b), \zeta_{\Gamma_{\aleph_j}^-}^-(\tau_b) = \sup_j \zeta_{\Gamma_{\aleph_j}^-}^-(\tau_b), \\ \Lambda_{\aleph_j}^+(\tau_b) = \sup_j \Lambda_{\aleph_j}^+(\tau_b), \zeta_{\Lambda_{\aleph_j}^+}^+(\tau_b) = \sup_j \zeta_{\Lambda_{\aleph_j}^+}^+(\tau_b), \Lambda_{\aleph_j}^-(\tau_b) = \inf_j \Lambda_{\aleph_j}^-(\tau_b), \zeta_{\Lambda_{\aleph_j}^-}^-(\tau_b) = \inf_j \zeta_{\Lambda_{\aleph_j}^-}^-(\tau_b), \\ \Phi_{\aleph_j}^+(\tau_b) = \inf_j \Phi_{\aleph_j}^+(\tau_b), \zeta_{\Phi_{\aleph_j}^+}^+(\tau_b) = \inf_j \zeta_{\Phi_{\aleph_j}^+}^+(\tau_b), \Phi_{\aleph_j}^-(\tau_b) = \sup_j \Phi_{\aleph_j}^-(\tau_b), \zeta_{\Phi_{\aleph_j}^-}^-(\tau_b) = \sup_j \zeta_{\Phi_{\aleph_j}^-}^-(\tau_b), \\ \Psi_{\aleph_j}^+(\tau_b) = \sup_j \Psi_{\aleph_j}^+(\tau_b), \zeta_{\Psi_{\aleph_j}^+}^+(\tau_b) = \sup_j \zeta_{\Psi_{\aleph_j}^+}^+(\tau_b), \Psi_{\aleph_j}^-(\tau_b) = \inf_j \Psi_{\aleph_j}^-(\tau_b), \zeta_{\Psi_{\aleph_j}^-}^-(\tau_b) = \inf_j \zeta_{\Psi_{\aleph_j}^-}^-(\tau_b) \end{array} \right\}_{(b \in \mathbb{N})}$$

Example 3.2. Assume $\aleph_j, j = 1, 2, 3$ be the BCLDFS in the reference set $\Delta = \{\tau_1, \tau_2\}$ where,

$$\aleph_1 = \{(\tau_1, \langle 0.7e^{i2\pi(0.8)}, -0.9e^{i2\pi(-0.7)}, 0.4e^{i2\pi(0.5)}, -0.2e^{i2\pi(-0.1)} \rangle, \langle 0.8e^{i2\pi(0.9)}, -0.7e^{i2\pi(-0.6)}, 0.2e^{i2\pi(0.1)}, -0.3e^{i2\pi(-0.1)} \rangle), (\tau_2, \langle 0.6e^{i2\pi(0.9)}, -0.9e^{i2\pi(-0.6)}, 0.5e^{i2\pi(0.7)}, -0.3e^{i2\pi(-0.4)} \rangle, \langle 0.7e^{i2\pi(0.5)}, -0.4e^{i2\pi(-0.3)}, 0.2e^{i2\pi(0.4)}, -0.4e^{i2\pi(-0.2)} \rangle))\}$$

$$\aleph_2 = \{(\tau_1, \langle 0.6e^{i2\pi(0.7)}, -0.8e^{i2\pi(-0.6)}, 0.5e^{i2\pi(0.8)}, -0.6e^{i2\pi(-0.5)} \rangle, \langle 0.7e^{i2\pi(0.4)}, -0.4e^{i2\pi(-0.6)}, 0.3e^{i2\pi(0.5)}, -0.2e^{i2\pi(-0.3)} \rangle), (\tau_2, \langle 0.5e^{i2\pi(0.6)}, -0.9e^{i2\pi(-0.8)}, 0.4e^{i2\pi(0.8)}, -0.6e^{i2\pi(-0.5)} \rangle, \langle 0.7e^{i2\pi(0.4)}, -0.5e^{i2\pi(-0.4)}, 0.1e^{i2\pi(0.5)}, -0.4e^{i2\pi(-0.1)} \rangle))\}$$

$$\aleph_3 = \{(\tau_1, \langle 0.3e^{i2\pi(0.2)}, -0.2e^{i2\pi(-0.1)}, 0.5e^{i2\pi(0.7)}, -0.8e^{i2\pi(-0.9)} \rangle, \langle 0.4e^{i2\pi(0.6)}, -0.6e^{i2\pi(-0.5)}, 0.3e^{i2\pi(0.4)}, -0.4e^{i2\pi(-0.5)} \rangle), (\tau_2, \langle 0.4e^{i2\pi(0.4)}, -0.6e^{i2\pi(-0.8)}, 0.7e^{i2\pi(0.5)}, -0.9e^{i2\pi(-0.1)} \rangle, \langle 0.4e^{i2\pi(0.6)}, -0.5e^{i2\pi(-0.5)}, 0.5e^{i2\pi(0.3)}, -0.4e^{i2\pi(-0.3)} \rangle))\}. \text{ Then,}$$

- $\aleph_1^c = \{(\tau_1, \langle 0.4e^{i2\pi(0.5)}, -0.2e^{i2\pi(-0.1)}, 0.7e^{i2\pi(0.8)}, -0.9e^{i2\pi(-0.7)} \rangle, \langle 0.2e^{i2\pi(0.1)}, -0.3e^{i2\pi(-0.1)}, 0.8e^{i2\pi(0.9)}, -0.7e^{i2\pi(-0.6)} \rangle), (\tau_2, \langle 0.5e^{i2\pi(0.7)}, -0.3e^{i2\pi(-0.4)}, 0.6e^{i2\pi(0.9)}, -0.9e^{i2\pi(-0.6)} \rangle, \langle 0.2e^{i2\pi(0.4)}, -0.4e^{i2\pi(-0.2)}, 0.7e^{i2\pi(0.5)}, -0.4e^{i2\pi(-0.3)} \rangle))\}$

- $\cup_{j=1}^3 \aleph_j = \{(\tau_1, \langle 0.7e^{i2\pi(0.8)}, -0.9e^{i2\pi(-0.7)}, 0.4e^{i2\pi(0.5)}, -0.2e^{i2\pi(-0.1)} \rangle, \langle 0.8e^{i2\pi(0.9)}, -0.7e^{i2\pi(-0.6)}, 0.2e^{i2\pi(0.1)}, -0.2e^{i2\pi(-0.1)} \rangle), (\tau_2, \langle 0.6e^{i2\pi(0.9)}, -0.9e^{i2\pi(-0.8)}, 0.4e^{i2\pi(0.5)}, -0.3e^{i2\pi(-0.1)} \rangle, \langle 0.7e^{i2\pi(0.6)}, -0.5e^{i2\pi(-0.5)}, 0.1e^{i2\pi(0.3)}, -0.4e^{i2\pi(-0.3)} \rangle))\}$

- $\cap_{j=1}^3 \aleph_j = \{(\tau_1, \langle 0.3e^{i2\pi(0.2)}, -0.2e^{i2\pi(-0.1)}, 0.5e^{i2\pi(0.7)}, -0.8e^{i2\pi(-0.9)} \rangle, \langle 0.4e^{i2\pi(0.4)}, -0.4e^{i2\pi(-0.5)}, 0.3e^{i2\pi(0.5)}, -0.4e^{i2\pi(-0.5)} \rangle), (\tau_2, \langle 0.4e^{i2\pi(0.4)}, -0.6e^{i2\pi(-0.6)}, 0.7e^{i2\pi(0.8)}, -0.9e^{i2\pi(-0.5)} \rangle, \langle 0.4e^{i2\pi(0.4)}, -0.4e^{i2\pi(-0.3)}, 0.5e^{i2\pi(0.5)}, -0.4e^{i2\pi(-0.1)} \rangle))\}$

Proposition 3.2. If $\mathfrak{B}_j \in BCLDFS(\Delta)$ with $1 \leq j \leq n$, then $\cup_{j=1}^n \mathfrak{B}_j$ and $\cap_{j=1}^n \mathfrak{B}_j$ forms a BCLDFS in Δ .

Proposition 3.3. Assume $\mathfrak{A}_j, j = 1, 2, 3$ in BCLDFS(Δ). Then the following hold

(i) $\mathfrak{A}_1 \cup \mathfrak{A}_2 = \mathfrak{A}_2 \cup \mathfrak{A}_1$

(ii) $\mathfrak{A}_1 \cup (\mathfrak{A}_2 \cup \mathfrak{A}_3) = (\mathfrak{A}_1 \cup \mathfrak{A}_2) \cup \mathfrak{A}_3$

(iii) $\mathfrak{A}_1 \cup (\mathfrak{A}_2 \cap \mathfrak{A}_3) = (\mathfrak{A}_1 \cup \mathfrak{A}_2) \cap (\mathfrak{A}_1 \cup \mathfrak{A}_3)$

$$(iv) (\mathfrak{A}_1 \cup \mathfrak{A}_2)^c = \mathfrak{A}_1^c \cap \mathfrak{A}_2^c$$

Proof. The proof follows directly from the definition 3.3. $\square$

3.2 Operational Laws and their characteristics on BCLDFNs

Definition 3.4. For two BCLDFNs $\mathfrak{d}_j = (\langle (\Gamma_{\mathfrak{N}_j}^+, \zeta_{\Gamma_{\mathfrak{N}_j}}^+), (\Gamma_{\mathfrak{N}_j}^-, \zeta_{\Gamma_{\mathfrak{N}_j}}^-), (\Lambda_{\mathfrak{N}_j}^+, \zeta_{\Lambda_{\mathfrak{N}_j}}^+), (\Lambda_{\mathfrak{N}_j}^-, \zeta_{\Lambda_{\mathfrak{N}_j}}^-) \rangle, \langle (\Phi_{\mathfrak{N}_j}^+, \zeta_{\Phi_{\mathfrak{N}_j}}^+), (\Phi_{\mathfrak{N}_j}^-, \zeta_{\Phi_{\mathfrak{N}_j}}^-), (\Psi_{\mathfrak{N}_j}^+, \zeta_{\Psi_{\mathfrak{N}_j}}^+), (\Psi_{\mathfrak{N}_j}^-, \zeta_{\Psi_{\mathfrak{N}_j}}^-) \rangle), j = 1, 2$ with $\kappa > 0$ we define

$$\begin{aligned} \bullet \mathfrak{d}_1 \oplus \mathfrak{d}_2 = & (\langle (\Gamma_{\mathfrak{N}_1}^+ + \Gamma_{\mathfrak{N}_2}^+ - \Gamma_{\mathfrak{N}_1}^+ \Gamma_{\mathfrak{N}_2}^+, \zeta_{\Gamma_{\mathfrak{N}_1}}^+ + \zeta_{\Gamma_{\mathfrak{N}_2}}^+ - \zeta_{\Gamma_{\mathfrak{N}_1}}^+ \zeta_{\Gamma_{\mathfrak{N}_2}}^+), ((-\Gamma_{\mathfrak{N}_1}^-)(-\Gamma_{\mathfrak{N}_2}^-)), -((-\zeta_{\Gamma_{\mathfrak{N}_1}}^-)(-\zeta_{\Gamma_{\mathfrak{N}_2}}^-)), \\ & (\Lambda_{\mathfrak{N}_1}^+ \Lambda_{\mathfrak{N}_2}^+, \zeta_{\Lambda_{\mathfrak{N}_1}}^+ \zeta_{\Lambda_{\mathfrak{N}_2}}^+), -((-\Lambda_{\mathfrak{N}_1}^-)(-\Lambda_{\mathfrak{N}_2}^-)), -((-\zeta_{\Lambda_{\mathfrak{N}_1}}^-)(-\zeta_{\Lambda_{\mathfrak{N}_2}}^-)), \\ & \langle (\Phi_{\mathfrak{N}_1}^+ + \Phi_{\mathfrak{N}_2}^+ - \Phi_{\mathfrak{N}_1}^+ \Phi_{\mathfrak{N}_2}^+, \zeta_{\Phi_{\mathfrak{N}_1}}^+ + \zeta_{\Phi_{\mathfrak{N}_2}}^+ - \zeta_{\Phi_{\mathfrak{N}_1}}^+ \zeta_{\Phi_{\mathfrak{N}_2}}^+), -((-\Phi_{\mathfrak{N}_1}^-)(-\Phi_{\mathfrak{N}_2}^-)), -((-\zeta_{\Phi_{\mathfrak{N}_1}}^-)(-\zeta_{\Phi_{\mathfrak{N}_2}}^-)), \\ & (\Psi_{\mathfrak{N}_1}^+ \Psi_{\mathfrak{N}_2}^+, \zeta_{\Psi_{\mathfrak{N}_1}}^+ \zeta_{\Psi_{\mathfrak{N}_2}}^+), -((-\Psi_{\mathfrak{N}_1}^-)(-\Psi_{\mathfrak{N}_2}^-)), -((-\zeta_{\Psi_{\mathfrak{N}_1}}^-)(-\zeta_{\Psi_{\mathfrak{N}_2}}^-)) \rangle \rangle \end{aligned}$$

$$\begin{aligned} \bullet \mathfrak{d}_1 \otimes \mathfrak{d}_2 = & (\langle (\Gamma_{\mathfrak{N}_1}^+ \Gamma_{\mathfrak{N}_2}^+, \zeta_{\Gamma_{\mathfrak{N}_1}}^+ \zeta_{\Gamma_{\mathfrak{N}_2}}^+), -((-\Gamma_{\mathfrak{N}_1}^-)(-\Gamma_{\mathfrak{N}_2}^-)), -((-\zeta_{\Gamma_{\mathfrak{N}_1}}^-)(-\zeta_{\Gamma_{\mathfrak{N}_2}}^-)), \\ & (\Lambda_{\mathfrak{N}_1}^+ \Lambda_{\mathfrak{N}_2}^+, \zeta_{\Lambda_{\mathfrak{N}_1}}^+ \zeta_{\Lambda_{\mathfrak{N}_2}}^+), -((-\Lambda_{\mathfrak{N}_1}^-)(-\Lambda_{\mathfrak{N}_2}^-)), -((-\zeta_{\Lambda_{\mathfrak{N}_1}}^-)(-\zeta_{\Lambda_{\mathfrak{N}_2}}^-)), \\ & \langle (\Phi_{\mathfrak{N}_1}^+ \Phi_{\mathfrak{N}_2}^+, \zeta_{\Phi_{\mathfrak{N}_1}}^+ \zeta_{\Phi_{\mathfrak{N}_2}}^+), -((-\Phi_{\mathfrak{N}_1}^-)(-\Phi_{\mathfrak{N}_2}^-)), -((-\zeta_{\Phi_{\mathfrak{N}_1}}^-)(-\zeta_{\Phi_{\mathfrak{N}_2}}^-)), \\ & (\Psi_{\mathfrak{N}_1}^+ \Psi_{\mathfrak{N}_2}^+, \zeta_{\Psi_{\mathfrak{N}_1}}^+ \zeta_{\Psi_{\mathfrak{N}_2}}^+), -((-\Psi_{\mathfrak{N}_1}^-)(-\Psi_{\mathfrak{N}_2}^-)), -((-\zeta_{\Psi_{\mathfrak{N}_1}}^-)(-\zeta_{\Psi_{\mathfrak{N}_2}}^-)) \rangle \rangle \end{aligned}$$

$$\begin{aligned} \bullet \kappa \mathfrak{d}_1 = & (\langle (1 - (1 - \Gamma_{\mathfrak{N}_1}^+)^{\kappa}), 1 - (1 - \zeta_{\Gamma_{\mathfrak{N}_1}}^+)^{\kappa}), -((-\Gamma_{\mathfrak{N}_1}^-)^{\kappa}), -((-\zeta_{\Gamma_{\mathfrak{N}_1}}^-)^{\kappa}), ((\Lambda_{\mathfrak{N}_1}^+)^{\kappa}), (\zeta_{\Lambda_{\mathfrak{N}_1}}^+)^{\kappa}), \\ & (- (1 - (1 - (-\Lambda_{\mathfrak{N}_1}^-)^{\kappa})), - (1 - (1 - (-\zeta_{\Lambda_{\mathfrak{N}_1}}^-)^{\kappa}))), \langle (1 - (1 - \Phi_{\mathfrak{N}_1}^+)^{\kappa}), 1 - (1 - \zeta_{\Phi_{\mathfrak{N}_1}}^+)^{\kappa}), -((-\Phi_{\mathfrak{N}_1}^-)^{\kappa}), \\ & (- (1 - (1 - (-\zeta_{\Phi_{\mathfrak{N}_1}}^-)^{\kappa}))), ((\Psi_{\mathfrak{N}_1}^+)^{\kappa}), (\zeta_{\Psi_{\mathfrak{N}_1}}^+)^{\kappa}), - (1 - (1 - (-\Psi_{\mathfrak{N}_1}^-)^{\kappa})), - (1 - (1 - (-\zeta_{\Psi_{\mathfrak{N}_1}}^-)^{\kappa}))) \rangle \rangle \end{aligned}$$

$$\begin{aligned} \bullet \mathfrak{d}_1^{\kappa} = & (\langle ((\Gamma_{\mathfrak{N}_1}^+)^{\kappa}), (\zeta_{\Gamma_{\mathfrak{N}_1}}^+)^{\kappa}), - (1 - (1 - (-\Gamma_{\mathfrak{N}_1}^-)^{\kappa})), - (1 - (1 - (-\zeta_{\Gamma_{\mathfrak{N}_1}}^-)^{\kappa}))), (1 - (1 - \Lambda_{\mathfrak{N}_1}^+)^{\kappa}), 1 - \\ & (1 - \zeta_{\Lambda_{\mathfrak{N}_1}}^+)^{\kappa}), -((-\Lambda_{\mathfrak{N}_1}^-)^{\kappa}), -((-\zeta_{\Lambda_{\mathfrak{N}_1}}^-)^{\kappa})), \langle ((\Phi_{\mathfrak{N}_1}^+)^{\kappa}), (\zeta_{\Phi_{\mathfrak{N}_1}}^+)^{\kappa}), - (1 - (1 - (-\Phi_{\mathfrak{N}_1}^-)^{\kappa})), - (1 - \\ & (1 - (-\zeta_{\Phi_{\mathfrak{N}_1}}^-)^{\kappa}))), (1 - (1 - \Psi_{\mathfrak{N}_1}^+)^{\kappa}), 1 - (1 - \zeta_{\Psi_{\mathfrak{N}_1}}^+)^{\kappa}), -((-\Psi_{\mathfrak{N}_1}^-)^{\kappa}), -((-\zeta_{\Psi_{\mathfrak{N}_1}}^-)^{\kappa})) \rangle \rangle \end{aligned}$$

Example 3.3. Let $\mathfrak{d}_1 = (\langle 0.7e^{i2\pi(0.8)}, -0.9e^{i2\pi(-0.7)}, 0.4e^{i2\pi(0.5)}, -0.2e^{i2\pi(-0.1)} \rangle, \langle 0.8e^{i2\pi(0.9)}, -0.7e^{i2\pi(-0.6)}, 0.2e^{i2\pi(0.1)}, -0.3e^{i2\pi(-0.1)} \rangle)$ and $\mathfrak{d}_2 = (\langle 0.6e^{i2\pi(0.9)}, -0.9e^{i2\pi(-0.6)}, 0.5e^{i2\pi(0.7)}, -0.3e^{i2\pi(-0.4)} \rangle, \langle 0.7e^{i2\pi(0.5)}, -0.4e^{i2\pi(-0.3)}, 0.2e^{i2\pi(0.4)}, -0.4e^{i2\pi(-0.2)} \rangle)$ be two BCLDFNs. Then

$$\bullet \mathfrak{d}_1 \oplus \mathfrak{d}_2 = (\langle 0.88e^{i2\pi(0.98)}, -0.81e^{i2\pi(-0.42)}, 0.20e^{i2\pi(0.35)}, -0.44e^{i2\pi(-0.46)} \rangle, \langle 0.94e^{i2\pi(0.95)}, -0.28e^{i2\pi(-0.18)}, 0.04e^{i2\pi(0.04)}, -0.58e^{i2\pi(-0.28)} \rangle)$$

$$\bullet \mathfrak{d}_1 \otimes \mathfrak{d}_2 = (\langle 0.42e^{i2\pi(0.72)}, -0.99e^{i2\pi(-0.88)}, 0.70e^{i2\pi(0.85)}, -0.06e^{i2\pi(-0.04)} \rangle, \langle 0.56e^{i2\pi(0.45)}, -0.82e^{i2\pi(-0.72)}, 0.36e^{i2\pi(0.46)}, -0.12e^{i2\pi(-0.02)} \rangle)$$

For $\kappa = 0.5$,

- $\kappa \mathfrak{d}_1 = (\langle 0.4523e^{i2\pi(0.5528)}, -0.9487e^{i2\pi(-0.8367)}, 0.6325e^{i2\pi(0.7071)}, -0.1056e^{i2\pi(-0.0513)} \rangle, \langle 0.5528e^{i2\pi(0.6838)}, -0.8367e^{i2\pi(-0.7746)}, 0.4472e^{i2\pi(0.3162)}, -0.1633e^{i2\pi(-0.0513)} \rangle)$
- $\mathfrak{d}_1^\kappa = (\langle 0.8367e^{i2\pi(0.8944)}, -0.6838e^{i2\pi(-0.4523)}, 0.2254e^{i2\pi(0.2929)}, -0.4472e^{i2\pi(-0.3162)} \rangle, \langle 0.8944e^{i2\pi(0.9487)}, -0.4523e^{i2\pi(-0.3676)}, 0.1056e^{i2\pi(0.0513)}, -0.5477e^{i2\pi(-0.3162)} \rangle)$

Proposition 3.4. If $\mathfrak{d}_1$ and $\mathfrak{d}_2$ are two BCLDFNs and $\kappa, \kappa_1, \kappa_2 > 0$, then $\mathfrak{d}_1 \oplus \mathfrak{d}_2$, $\mathfrak{d}_1 \otimes \mathfrak{d}_2$, $\kappa \mathfrak{d}_1$ and $\mathfrak{d}_1^\kappa$ are also BCLDFNs and the following properties hold

$$(i) \quad \mathfrak{d}_1 \oplus \mathfrak{d}_2 = \mathfrak{d}_2 \oplus \mathfrak{d}_1$$

$$(ii) \quad \mathfrak{d}_1 \otimes \mathfrak{d}_2 = \mathfrak{d}_2 \otimes \mathfrak{d}_1$$

$$(iii) \quad \kappa(\mathfrak{d}_1 \oplus \mathfrak{d}_2) = \kappa \mathfrak{d}_1 \oplus \kappa \mathfrak{d}_2$$

$$(iv) \quad (\mathfrak{d}_1 \otimes \mathfrak{d}_2)^\kappa = \mathfrak{d}_1^\kappa \otimes \mathfrak{d}_2^\kappa$$

$$(v) \quad (\kappa_1 + \kappa_2) \mathfrak{d}_1 = \kappa_1 \mathfrak{d}_1 + \kappa_2 \mathfrak{d}_2$$

$$(vi) \quad \mathfrak{d}_1^{\kappa_1 + \kappa_2} = \mathfrak{d}_1^{\kappa_1} \otimes \mathfrak{d}_1^{\kappa_2}$$

Proof. The proof is straightforward from the definition 3.4 and is similar to that in [82]. $\square$

Theorem 3.1. Suppose let $\mathfrak{d}_j = (\langle (\Gamma_{\mathfrak{N}_j}^+, \zeta_{\Gamma_{\mathfrak{N}_j}}^+), (\Gamma_{\mathfrak{N}_j}^-, \zeta_{\Gamma_{\mathfrak{N}_j}}^-), (\Lambda_{\mathfrak{N}_j}^+, \zeta_{\Lambda_{\mathfrak{N}_j}}^+), (\Lambda_{\mathfrak{N}_j}^-, \zeta_{\Lambda_{\mathfrak{N}_j}}^-) \rangle, \langle (\Phi_{\mathfrak{N}_j}^+, \zeta_{\Phi_{\mathfrak{N}_j}}^+), (\Phi_{\mathfrak{N}_j}^-, \zeta_{\Phi_{\mathfrak{N}_j}}^-), (\Psi_{\mathfrak{N}_j}^+, \zeta_{\Psi_{\mathfrak{N}_j}}^+), (\Psi_{\mathfrak{N}_j}^-, \zeta_{\Psi_{\mathfrak{N}_j}}^-) \rangle)$, $1 \leq j \leq n$ be a family of BCLDFNs on Δ and $\rho_j = (\rho_1, \dots, \rho_n)^T$ be the weight vectors associated to each $\mathfrak{d}_j$ with each $\rho_j > 0$ and $\sum_{j=1}^n \rho_j = 1$. Then bipolar complex linear diophantine fuzzy weighted geometric aggregation operator (BCLDFWGAO) is a mapping $F : BCLDFN(\Delta) \rightarrow BCLDFN(\Delta)$ defined as

$$BCLDFWGAO(\mathfrak{d}_1, \dots, \mathfrak{d}_n) = \prod_{j=1}^n (\mathfrak{d}_j)^{\rho_j}$$

$$= \left(\left\langle \left(\prod_{j=1}^n (\Gamma_{\mathfrak{N}_j}^+)^{\rho_j}, \prod_{j=1}^n (\zeta_{\Gamma_{\mathfrak{N}_j}}^+)^{\rho_j} \right), \left(- \left(1 - \prod_{j=1}^n (1 - (-\Gamma_{\mathfrak{N}_j}^-)^{\rho_j}) \right)^{\rho_j} \right), \left(1 - \prod_{j=1}^n (1 - \Lambda_{\mathfrak{N}_j}^+)^{\rho_j}, 1 - \prod_{j=1}^n (1 - \zeta_{\Lambda_{\mathfrak{N}_j}}^+)^{\rho_j} \right), \right. \right. \\ \left. \left(- \left(\prod_{j=1}^n (-\Lambda_{\mathfrak{N}_j}^-)^{\rho_j} \right), - \left(\prod_{j=1}^n (-\zeta_{\Lambda_{\mathfrak{N}_j}}^-)^{\rho_j} \right) \right), \left\langle \left(\prod_{j=1}^n (\Phi_{\mathfrak{N}_j}^+)^{\rho_j}, \prod_{j=1}^n (\zeta_{\Phi_{\mathfrak{N}_j}}^+)^{\rho_j} \right), \right. \right. \\ \left. \left(- \left(1 - \prod_{j=1}^n (1 - (-\Phi_{\mathfrak{N}_j}^-)^{\rho_j}) \right)^{\rho_j} \right), - \left(1 - \prod_{j=1}^n (1 - (-\zeta_{\Phi_{\mathfrak{N}_j}}^-)^{\rho_j}) \right), \left(1 - \prod_{j=1}^n (1 - \Psi_{\mathfrak{N}_j}^+)^{\rho_j}, \right. \right. \\ \left. \left. 1 - \prod_{j=1}^n (1 - \zeta_{\Psi_{\mathfrak{N}_j}}^+)^{\rho_j} \right), \left(- \left(\prod_{j=1}^n (-\Psi_{\mathfrak{N}_j}^-)^{\rho_j} \right), - \left(\prod_{j=1}^n (-\zeta_{\Psi_{\mathfrak{N}_j}}^-)^{\rho_j} \right) \right) \right\rangle \right) \quad (?)$$

Proof. The proof follows, utilizing the operational rules established in definition 3.4 and following mathematical induction [82]. $\square$

371 **Definition 3.5.** For a BCLDFN in F , $\mathcal{H} = (\langle(\Gamma_{\mathcal{N}}^+, \zeta_{\Gamma_{\mathcal{N}}}^+), (\Gamma_{\mathcal{N}}^-, \zeta_{\Gamma_{\mathcal{N}}}^-), (\Lambda_{\mathcal{N}_j}^+, \zeta_{\Lambda_{\mathcal{N}}}^+), (\Lambda_{\mathcal{N}}^-, \zeta_{\Lambda_{\mathcal{N}}}^-)\rangle,$
 372 $\langle(\Phi_{\mathcal{N}}^+, \zeta_{\Phi_{\mathcal{N}}}^+), (\Phi_{\mathcal{N}}^-, \zeta_{\Phi_{\mathcal{N}}}^-), (\Psi_{\mathcal{N}}^+, \zeta_{\Psi_{\mathcal{N}}}^+), (\Psi_{\mathcal{N}}^-, \zeta_{\Psi_{\mathcal{N}}}^-)\rangle)$ we define the following

373 • The score function χ is a mapping $\chi : BCLDFN(\Delta) \rightarrow [-1, 1]$ defined as

$$\chi(\mathcal{H}_1) = \frac{1}{8} \left[\left\{ (\Gamma_{\mathcal{N}}^+ - \Lambda_{\mathcal{N}}^+) + (\zeta_{\Gamma_{\mathcal{N}}}^+ - \zeta_{\Lambda_{\mathcal{N}}}^+) + (\Phi_{\mathcal{N}}^+ - \Psi_{\mathcal{N}}^+) + (\zeta_{\Phi_{\mathcal{N}}}^+ - \zeta_{\Psi_{\mathcal{N}}}^+) \right\} - \right. \\ \left. \left\{ (\Gamma_{\mathcal{N}}^- - \Lambda_{\mathcal{N}}^-) + (\zeta_{\Gamma_{\mathcal{N}}}^- - \zeta_{\Lambda_{\mathcal{N}}}^-) + (\Phi_{\mathcal{N}}^- - \Psi_{\mathcal{N}}^-) + (\zeta_{\Phi_{\mathcal{N}}}^- - \zeta_{\Psi_{\mathcal{N}}}^-) \right\} \right] \quad (3)$$

374 • The accuracy function ξ is a mapping $\xi : BCLDFN(\Delta) \rightarrow [0, 1]$ defined as

$$\xi(\mathcal{H}) = \frac{1}{8} \left[\left\{ \left(\frac{\Gamma_{\mathcal{N}}^+ + \Lambda_{\mathcal{N}}^+}{2} \right) + \left(\frac{\zeta_{\Gamma_{\mathcal{N}}}^+ + \zeta_{\Lambda_{\mathcal{N}}}^+}{2} \right) + (\Phi_{\mathcal{N}}^+ + \Psi_{\mathcal{N}}^+) + (\zeta_{\Phi_{\mathcal{N}}}^+ + \zeta_{\Psi_{\mathcal{N}}}^+) \right\} - \right. \\ \left. \left\{ \left(\frac{\Gamma_{\mathcal{N}}^- + \Lambda_{\mathcal{N}}^-}{2} \right) + \left(\frac{\zeta_{\Gamma_{\mathcal{N}}}^- + \zeta_{\Lambda_{\mathcal{N}}}^-}{2} \right) + (\Phi_{\mathcal{N}}^- + \Psi_{\mathcal{N}}^-) + (\zeta_{\Phi_{\mathcal{N}}}^- + \zeta_{\Psi_{\mathcal{N}}}^-) \right\} \right] \quad (4)$$

375 The two BCLDFNs $\mathfrak{d}_1$ and $\mathfrak{d}_2$ are contrasted using the eqns (3) and (4) utilising the compar-
 376 ison procedure stated below.

377 1. $\chi(\mathfrak{d}_1) > \chi(\mathfrak{d}_2) \implies \mathfrak{d}_1 > \mathfrak{d}_2$

378 2. $\chi(\mathfrak{d}_1) < \chi(\mathfrak{d}_2) \implies \mathfrak{d}_1 < \mathfrak{d}_2$

379 3. $\chi(\mathfrak{d}_1) = \chi(\mathfrak{d}_2)$, then

380 (a) $\xi(\mathfrak{d}_1) > \xi(\mathfrak{d}_2) \implies \mathfrak{d}_1 > \mathfrak{d}_2$

381 (b) $\xi(\mathfrak{d}_1) < \xi(\mathfrak{d}_2) \implies \mathfrak{d}_1 < \mathfrak{d}_2$

382 (c) $\xi(\mathfrak{d}_1) = \xi(\mathfrak{d}_2) \implies \mathfrak{d}_1 = \mathfrak{d}_2$

383 4 Methodology

384 The study proposes a sophisticated decision model for ranking renewable energy sources
 385 for hydrogen generation in India. The combined framework incorporates the LBWA model
 386 for assessing significant criteria and an improved TRUST approach for prioritising alterna-
 387 tives using the BCLDFS. The figure 2 demonstrates the suggested framework's hierarchical
 388 structure.

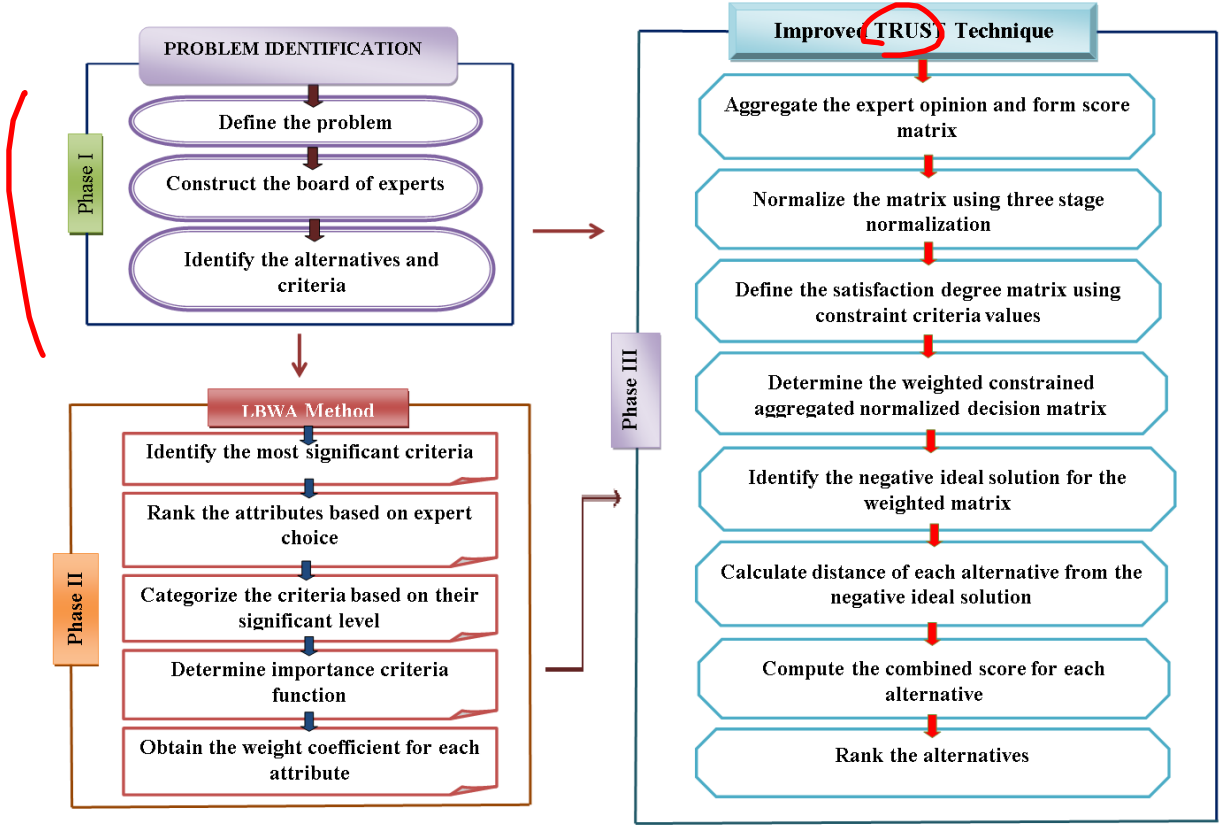


Figure 2: Structure of the proposed model

4.1 Problem definition

A fuzzy MAGDM problem under investigation consists of a collection of l alternatives $\mathfrak{C}_i, (1 \leq i \leq l)$ that need to be appraised based on predetermined set of attributes $\mathfrak{h} = \{\mathfrak{A}_f : 1 \leq k \leq f\}$ with $\mathfrak{h}^+$ and $\mathfrak{h}^-$ representing the set of positive and negative constraints. Let v_k represent the weight vectors associated to each $\mathfrak{A}_k$, presuming that $\sum_{k=1}^f v_k = 1$ and $v_k > 0$ are true. In order to discover the best option, a committee of m experts $\mathfrak{P}_e, (1 \leq e \leq m)$ is convened to look into the situation and provide input on each alternatives in terms of each attribute. The impact of each expert in the process of making decisions is determined by the accompanying weight vectors δ_e , with $\delta_e > 0$ and $\sum_{e=1}^m \delta_e = 1$.

The performance rating provided for each alternatives against each attribute in the light

399 of each expert ($\mathfrak{P}_e$) is constructed as a fuzzy decision matrix taking the form:

$$\mathfrak{D}^e = (\mathfrak{o}_{ik}^e)_{l \times f} = \begin{matrix} & \mathfrak{J}_1 & \mathfrak{J}_2 & \cdots & \mathfrak{J}_f \\ \begin{matrix} \mathfrak{C}_1 \\ \mathfrak{C}_2 \\ \vdots \\ \mathfrak{C}_l \end{matrix} & \begin{pmatrix} \mathfrak{o}_{11}^e & \mathfrak{o}_{12}^e & \cdots & \mathfrak{o}_{1f}^e \\ \mathfrak{o}_{21}^e & \mathfrak{o}_{22}^e & \cdots & \mathfrak{o}_{2f}^e \\ \vdots & \vdots & \cdots & \vdots \\ \mathfrak{o}_{l1}^e & \mathfrak{o}_{l2}^e & \cdots & \mathfrak{o}_{lf}^e \end{pmatrix} \end{matrix}, \quad (5)$$

400 where each $\mathfrak{o}_{lf}^e$ is a BCLDFN expressed as

$$\mathfrak{o}_{lf}^e = \left(\begin{array}{c} \left\langle \left((\Gamma_{lf}^+)^e, (\zeta_{\Gamma_{lf}}^+)^e \right), \left((\Gamma_{lf}^-)^e, (\zeta_{\Gamma_{lf}}^-)^e \right), \left((\Lambda_{lf}^+)^e, (\zeta_{\Lambda_{lf}}^+)^e \right), \left((\Lambda_{lf}^-)^e, (\zeta_{\Lambda_{lf}}^-)^e \right) \right\rangle, \\ \left\langle \left((\Phi_{lf}^+)^e, (\zeta_{\Phi_{lf}}^+)^e \right), \left((\Phi_{lf}^-)^e, (\zeta_{\Phi_{lf}}^-)^e \right), \left((\Psi_{lf}^+)^e, (\zeta_{\Psi_{lf}}^+)^e \right), \left((\Psi_{lf}^-)^e, (\zeta_{\Psi_{lf}}^-)^e \right) \right\rangle \end{array} \right) \quad (6)$$

401 4.2 LBWA method

402 For the problem defined in section 4.1, the weight coefficients associated with each attribute
403 $\mathfrak{J}_k$ are obtained using the procedure outlined below:

404 **Step 1:** Identify the most significant criteria ($\hat{\mathfrak{J}}_k$) in terms of expert choice ($\mathfrak{P}_e$) from
405 the set $\mathfrak{h} = \{\mathfrak{J}_1, \mathfrak{J}_2, \dots, \mathfrak{J}_f\}$, and the rest are categorised into multiple degrees based on their
406 significance.

407 **Level $\mathfrak{T}_1$:** It includes criteria that are either equally or up to twice as essential as the
408 most influential parameter.

409 **Level $\mathfrak{T}_2$:** The attributes whose value is exactly twice as low as or up to three times
410 lower than the most significant criteria are included.

411

412 **Level $\mathfrak{T}_r$:** Similarly, criteria whose significance is r times less than or up to $r + 1$
413 times less than the most significant one are included.

414 As a result, a fundamental categorization of criteria in accordance with their degree of
415 relevance is generated. If $\mathfrak{T}(\Omega_j)$ indicates the significance level of the j^{th} criterion, then the
416 subsequent expression applies for any $\mathfrak{t} \in \{1, 2, \dots, r\}$:

$$\mathfrak{T} = \mathfrak{T}_1 \cup \mathfrak{T}_2 \cup \dots \mathfrak{T}_k \quad (7)$$

417 in addition to a requirement that if $\mathfrak{p}$, in $\{1, 2, \dots, r\}$ with $\mathfrak{p} \neq \mathfrak{q}$, then $\mathfrak{T}_p \cap \mathfrak{T}_q = \phi$.

418 **Step 2:** Each criterion $\mathfrak{J}_{t_p}$ belonging to $\mathfrak{T}_t = \{\mathfrak{J}_{t_1}, \mathfrak{J}_{t_2}, \dots, \mathfrak{J}_{t_h}\}$ is assigned an integer

value $\mathfrak{G}_{t_p}$ from the interval $[0, \alpha]$ where

$$\alpha = \max\{|\mathfrak{T}_1|, |\mathfrak{T}_2|, \dots, |\mathfrak{T}_r|\} \quad (8)$$

$\mathfrak{G}_1 = 0$ is substituted for $\hat{\mathfrak{J}}_f$ and for the others, the value is assigned based on their significance level. The criteria with more importance is assigned a comparatively larger integer value from $[0, \alpha]$ than the others.

Step 3: The value of α obtained by eqn (8) must meet $\alpha_0 > \alpha$, where α_0 is the elasticity coefficient.

Step 4: After each criterion has been assigned a significant value, the influence function $\mu : \mathfrak{T} \rightarrow \mathbb{R}$ for each criterion is determined using eqn (9) as

$$\mu(\mathfrak{J}_{t_p}) = \frac{\alpha_0}{t(\alpha_0) + \mathfrak{G}_{t_p}} \quad (9)$$

where t is the criterion level, α_0 is the elasticity coefficient and $\mathfrak{G}_{t_p} \in [0, \alpha]$.

Step 5: The weight coefficients for each criterion accordance to an expert $\mathfrak{P}_\epsilon$ are computed using eqn (10)

$$v_k^\epsilon = \begin{cases} \frac{1}{1 + \mu(\mathfrak{J}_k) + \dots + \mu(\mathfrak{J}_f)} & \text{for } \hat{\mathfrak{J}}_k \\ \mu(\mathfrak{J}_k) v_k^\epsilon & \text{for } \mathfrak{J}_k \neq \hat{\mathfrak{J}}_k \end{cases} \quad (10)$$

where v_k^ϵ is the weight coefficient of $\mathfrak{J}_k^\epsilon$ with respect to the expert $\mathfrak{P}_\epsilon$.

Step 6: The individual weights vectors are further combined using the geometric mean (GM) method as in eqn (11)

$$v_k = \sqrt[\epsilon]{(v_k^1 v_k^2 \dots v_k^\epsilon)} \quad (11)$$

where v_k is the combined weight coefficient of $\mathfrak{J}_k$ and ϵ is the number of experts.

4.3 Improved fuzzy TRUST method

Step 1: The individual decision matrices framed based on each expert's opinion are combined to form an aggregated decision matrix $\mathfrak{D}$. Further, using eqn (3), the score matrix obtained is denoted as $\mathfrak{S}$ which takes the form

$$\mathfrak{S} = [\mathfrak{s}_{ik}]_{I \times f} = \begin{pmatrix} \mathfrak{s}_{11} & \mathfrak{s}_{12} & \dots & \mathfrak{s}_{1f} \\ \mathfrak{s}_{21} & \mathfrak{s}_{22} & \dots & \mathfrak{s}_{2f} \\ \vdots & \vdots & \ddots & \vdots \\ \mathfrak{s}_{I1} & \mathfrak{s}_{I2} & \dots & \mathfrak{s}_{If} \end{pmatrix} \quad (12)$$

Step 2: Three stage normalization is performed to transform the score matrix into

440 normalized matrix :

441 **Step 2.1:** The linear ratio method

$$\mathfrak{s}_{ik}^{(1)} = \begin{cases} \frac{\mathfrak{s}_{ik}}{\max_i \mathfrak{s}_{ik}} & \text{for } k \in \mathfrak{h}^+ \\ \frac{\min_i \mathfrak{s}_{ik}}{\mathfrak{s}_{ik}} & \text{for } k \in \mathfrak{h}^- \end{cases} \quad (13)$$

442

443 **Step 2.2:** Linear sum method

$$\mathfrak{s}_{ik}^{(2)} = \begin{cases} \frac{\mathfrak{s}_{ik}}{\sum_{i=1}^I \mathfrak{s}_{ik}} & \text{for } k \in \mathfrak{h}^+ \\ \frac{\frac{1}{\mathfrak{s}_{ik}}}{\sum_{i=1}^I \frac{1}{\mathfrak{s}_{ik}}} & \text{for } k \in \mathfrak{h}^- \end{cases} \quad (14)$$

444

445 **Step 2.3:** Linear min-max method

$$\mathfrak{s}_{ik}^{(3)} = \begin{cases} \frac{\frac{\mathfrak{s}_{ik} - \min_i \mathfrak{s}_{ik}}{i}}{\max_i \mathfrak{s}_{ik} - \min_i \mathfrak{s}_{ik}} & \text{for } k \in \mathfrak{h}^+ \\ \frac{\frac{\max_i \mathfrak{s}_{ik} - \mathfrak{s}_{ik}}{i}}{\max_i \mathfrak{s}_{ik} - \min_i \mathfrak{s}_{ik}} & \text{for } k \in \mathfrak{h}^- \end{cases} \quad (15)$$

446

447 **Step 3:** The normalization matrices obtained in each step is aggregated to form the
448 aggregated normalized matrix using the eqn (16)

$$A_{ik} = \beta_1 \mathfrak{s}_{ik}^{(1)} + \beta_2 \mathfrak{s}_{ik}^{(2)} + \beta_3 \mathfrak{s}_{ik}^{(3)} \quad (16)$$

449 where β_i is a parameter corresponding to each normalization satisfying $\beta_1 + \beta_2 + \beta_3 = 1$

450 **Step 4:** Construction of a satisfaction degree matrix $Q = [Q_{ik}]_{I \times f}$ based on the constraint
451 value of the criterion

452 **Step 4.1:** From the entries of the matrix in eqn (12) determine the minimum and
453 the maximum values using $o_k^{min} = \min(\mathfrak{s}_{ik})$ and $o_k^{max} = \max(\mathfrak{s}_{ik})$. The constraint value $\mathfrak{L}$
454 with respect to each criterion is determined as

$$\mathfrak{L}_k = [LB_k, UB_k] \subseteq [o_k^{min}, o_k^{max}] \quad (17)$$

455 Here LB_k and UB_k represent the lower and upper bound of the constraint value of the
456 criterion which is determined based on the experts opinion or the technical feature of a
457 criterion and is limited between the values.

458 **Step 4.2:** The elements of the satisfaction degree matrix is obtained based on the

following equations (18)-(19)

For Positive Criterion:

$$Q_{ik} = \begin{cases} 1 & \text{if } \mathfrak{s}_{ik} \in [LB_k, UB_k] \\ 1 - \frac{LB_k - \mathfrak{s}_{ik}}{\text{Max}(LB_k - o_k^{\min}, o_k^{\max} - UB_k) + 1} & \text{if } \mathfrak{s}_{ik} \in [o_k^{\min}, LB_k] \\ 1 - \frac{1 - UB_k + \mathfrak{s}_{ik}}{\text{Max}(LB_k - o_k^{\min}, o_k^{\max} - UB_k) + 1} & \text{if } \mathfrak{s}_{ik} \in [UB_k, o_k^{\max}] \end{cases} \quad (18)$$

For Negative Criterion:

$$Q_{ik} = \begin{cases} \frac{1}{\text{Max}(LB_k - o_k^{\min}, o_k^{\max} - UB_k) + 1} & \text{if } \mathfrak{s}_{ik} \in [LB_k, UB_k] \\ \frac{LB_k - \mathfrak{s}_{ik}}{\text{Max}(LB_k - o_k^{\min}, o_k^{\max} - UB_k)} & \text{if } \mathfrak{s}_{ik} \in [o_k^{\min}, LB_k] \\ \frac{\mathfrak{s}_{ik} - UB_k}{\text{Max}(LB_k - o_k^{\min}, o_k^{\max} - UB_k)} & \text{if } \mathfrak{s}_{ik} \in [UB_k, o_k^{\max}] \end{cases} \quad (19)$$

Step 5: A constrained aggregated normalized decision matrix $\mathfrak{U} = [\mathfrak{U}_{ik}]_{I_f}$ is calculated using the expression in (20)

$$\mathfrak{U}_{ik} = \mathfrak{Q}_{ik} A_{ik} \quad (20)$$

Step 6: The entries of the matrix in eqn (20) are multiplied by the weight coefficients of each criterion to obtain the weighted matrix as in eqn (21)

$$\mathfrak{W}_{ik} = \mathfrak{U}_{ik} v_k \quad (21)$$

For the matrix as in above, the negative ideal solution corresponding to a criterion is determined based on eqn (22)

$$\varepsilon_k = \min_i \mathfrak{W}_{ik} \quad (22)$$

Step 7: The distance of the alternatives from the negative ideal solution is evaluated in two stages

Step 7.1: The normalized Euclidean and Manhattan distance measures are used to calculate the distance from the negative ideal as

$$\mathfrak{D}_i = \frac{1}{2f} \sqrt{\sum_{k=1}^f (\mathfrak{W}_{ik} - \varepsilon_k)^2} \quad (23)$$

$$\mathfrak{M}_i = \frac{1}{2f} \sum_{k=1}^f |\mathfrak{W}_{ik} - \varepsilon_k| \quad (24)$$

Step 7.2: The normalized Lorentzian and squared distance measure are determined

as in

$$\mathfrak{Z}_i = \frac{1}{2f} \sum_{k=1}^f \log(1 + |\mathfrak{W}_{ik} - \varepsilon_k|) \quad (25)$$

$$\mathfrak{N}_i = \frac{1}{2f} \sum_{k=1}^f \frac{(\mathfrak{W}_{ik} - \varepsilon_k)^2}{(\mathfrak{W}_{ik} + \varepsilon_k)} \quad (26)$$

Step 8: Using the distance measures, relative distance matrices $\mathfrak{D}\mathfrak{M} = [\varsigma_{ik}]_{l \times l}$ and

$\mathfrak{Z}\mathfrak{N} = [\lambda_{ik}]_{l \times l}$ are calculated as eqn (27)-(28)

$$\varsigma_{ir} = (\mathfrak{D}_i - \mathfrak{D}_r) + ((\mathfrak{D}_i - \mathfrak{D}_r) * (\mathfrak{M}_i - \mathfrak{M}_r)) \quad (27)$$

$$\varrho_{ir} = (\mathfrak{Z}_i - \mathfrak{Z}_r) + ((\mathfrak{Z}_i - \mathfrak{Z}_r) * (\mathfrak{N}_i - \mathfrak{N}_r)) \quad (28)$$

Step 9: A combined score value for each alternative is obtained as

$$\lambda_i = \gamma \sum_{r=1}^l \varsigma_{ir} + (1 - \gamma) \sum_{r=1}^l \varrho_{ir} \quad (29)$$

Here γ is a parameter assigned to be a non-zero value less than one. The combined score value thus obtained is used in prioritizing the alternatives. The alternative with the highest value is regarded as the optimal choice for the problem.

5 Case Study

The established decision framework is used to examine the realistic RES for H_2 production in India. A committee of three experts shares their perspectives on energy alternatives under specified limitations utilizing linguistic variables whose values are given in the form of the introduced novel BCLDFNs, as shown in Table 3. The analysis of five alternative energy options for H_2 generation is done in terms of technical, economic, social, political, and environmental aspects, along with 14 sub criteria. These options and standards were developed using pertinent information and professional judgement. The economic parameters are seen as negative standards, whereas the rest are regarded as positive factors. Fig 3 depicts the hierarchical nature of the considered problem. The supporting works for the determined criteria are shown in Table 4 together with their explanations.

Table 3: Linguistic variables and the BCLDF rating

Linguistic terms	Membership grades				Non-membership grades			
	Positive		Negative		Positive		Negative	
	Amplitude	Phase	Amplitude	Phase	Amplitude	Phase	Amplitude	Phase
Absolutely Important (AI)	0.99	0.83	-0.71	-0.88	0.25	0.15	-0.16	-0.24
Very Important (VI)	0.88	0.74	-0.61	-0.78	0.41	0.39	-0.26	-0.34
Moderately important (MI)	0.79	0.62	-0.51	-0.68	0.53	0.43	-0.36	-0.44
Equally Important (EI)	0.64	0.56	-0.41	-0.58	0.62	0.59	-0.46	-0.54
Moderately unimportant (MU)	0.55	0.47	-0.31	-0.48	0.71	0.62	-0.56	-0.64
Very unimportant (VU)	0.37	0.23	-0.21	-0.38	0.82	0.73	-0.66	-0.74
Absolutely unimportant (AU)	0.26	0.18	-0.11	-0.28	0.91	0.89	-0.86	-0.94

Reference constraints							
Positive		Negative		Positive		Negative	
Amplitude	Phase	Amplitude	Phase	Amplitude	Phase	Amplitude	Phase
0.90	0.80	-0.80	-0.75	0.10	0.03	-0.05	-0.07
0.70	0.70	-0.70	-0.65	0.30	0.10	-0.10	-0.15
0.60	0.50	-0.60	-0.55	0.40	0.20	-0.30	-0.35
0.50	0.40	-0.40	-0.45	0.50	0.40	-0.40	-0.45
0.40	0.20	-0.30	-0.35	0.60	0.50	-0.60	-0.55
0.30	0.10	-0.10	-0.15	0.70	0.70	-0.70	-0.65
0.10	0.03	-0.05	-0.07	0.90	0.80	-0.80	-0.75

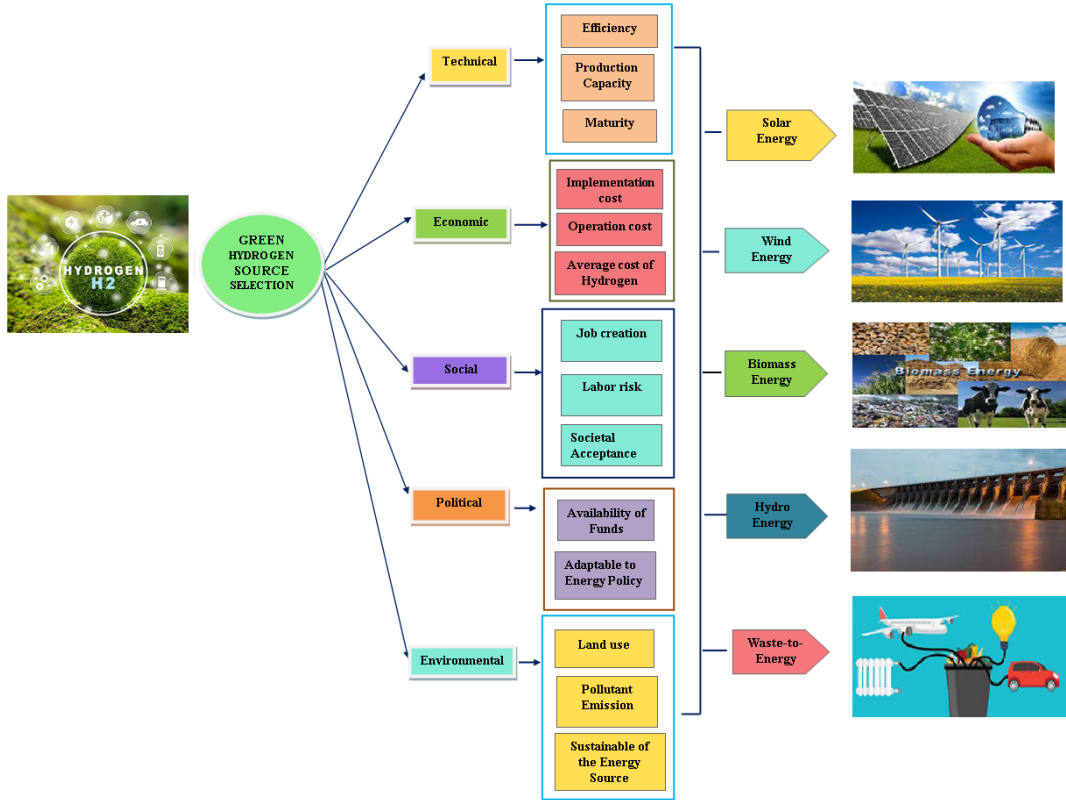


Figure 3: Green energy source selection problem

Table 4: Criteria explanation

Criteria	Sub-criteria	Description	References
Technical ($\mathfrak{C}_1$)	Efficiency ($\mathfrak{C}_{11}$)	It denotes the amount of H_2 generated from the input energy	Seddiki et al. [83]
	Production capacity ($\mathfrak{C}_{12}$)	It is the ability of the plant to produce the estimated amount of H_2 from the installed RE sources	Colak et al. [84]
	Maturity ($\mathfrak{C}_{13}$)	It alludes to the RE source's consistency for yielding H_2	Chen et al.[80], Colak et al. [84]
Economic ($\mathfrak{C}_2$)	Implementation cost ($\mathfrak{C}_{21}$)	It is the purchase cost including the equipment and installation cost	Seddiki et al. [83]
	Operational cost ($\mathfrak{C}_{22}$)	It includes amount incurred for the maintenance and operation of the equipment	Seddiki et al. [83]
	Average price of hydrogen ($\mathfrak{C}_{23}$)	It refers to the price of producing one unit of H_2 which is dependent on the operating and the maintenance cost	Abdel-Basset et al. [77], Colak et al. [84]
Social ($\mathfrak{C}_3$)	Job creation ($\mathfrak{C}_{31}$)	The installation of a plant should be able to create direct and indirect employment opportunities	Chen et al. [80]
	Labour risk ($\mathfrak{C}_{32}$)	It pertains to the danger involved in managing machinery and equipment.	Chen et al.[80]
	Societal Acceptance ($\mathfrak{C}_{33}$)	It reveals the residents' preferences towards RE plants.	Abdel-Basset et al. [77], Xu et al. [76]
Political ($\mathfrak{C}_4$)	Availability of funds ($\mathfrak{C}_{41}$)	It describes national and international sources of capital as well as government economic assistance	Colak et al.[84]
	Adaptable to the national energy policy ($\mathfrak{C}_{42}$)	It points to the potential for combining the RE source for H_2 production with the government's stated energy policy	Colak et al.[84]
Environmental ($\mathfrak{C}_5$)	Land use ($\mathfrak{C}_{51}$)	It signifies the surface area required for access to the production station	Chen et al.[80]
	Pollutant Emission ($\mathfrak{C}_{52}$)	It is a measure of hazardous gas emissions that lead to air pollution, and noise pollution during plant operation	Colak et al. [84]
	Sustainability of energy source ($\mathfrak{C}_{53}$)	It is the availability of an energy source that can be restored and used for a substantially greater amount of time.	Colak et al. [84]

Based on the criteria description, existing literature and experts standpoint $\mathfrak{C}_2$ is regarded as the negative criteria and the rest are assumed to be positive criteria.

5.1 Adaptation to the framework

5.1.1 Weight assessment of criteria and the sub criteria

Initially, the weight vectors of the main criteria are computed using the LBWA approach. The weights for each criterion in the considered set of five attributes are established by individual experts and merged using the geometric mean (GM) method. According to each expert, the criteria $\mathfrak{C}_1$, $\mathfrak{C}_3$, and $\mathfrak{C}_5$ are thought to be the most relevant. The following is a grouping of criteria based on the significance levels assigned by each expert ($\mathfrak{P}_e$):

$$\begin{array}{ccc}
 \mathfrak{P}_1 & \mathfrak{P}_2 & \mathfrak{P}_3 \\
 \mathfrak{T}_1 = \{\mathfrak{C}_1, \mathfrak{C}_5, \mathfrak{C}_3\} & \mathfrak{T}_1 = \{\mathfrak{C}_5, \mathfrak{C}_2, \mathfrak{C}_1\} & \mathfrak{T}_1 = \{\mathfrak{C}_3, \mathfrak{C}_5\} \\
 \mathfrak{T}_2 = \{\mathfrak{C}_4, \mathfrak{C}_2\} & \mathfrak{T}_2 = \{\mathfrak{C}_3, \mathfrak{C}_4\} & \mathfrak{T}_2 = \{\mathfrak{C}_1, \mathfrak{C}_2, \mathfrak{C}_4\}
 \end{array}$$

According to three experts, each criterion is compared on a scale of $[0, 3]$. The value of α_0 is set to 5 for $\mathfrak{P}_1$ and 4 for experts $\mathfrak{P}_2$ and $\mathfrak{P}_3$. Finally, the weight vectors for the main criteria are obtained as $v_k = (0.2141, 0.1479, 0.2013, 0.1255, 0.2698)$.

Table 5: Significance level for each sub criteria

Expert/ Criteria	$\mathfrak{P}_1$	Most relevant factor	$\mathfrak{P}_2$	Most relevant factor	$\mathfrak{P}_3$	Most relevant factor
$\mathfrak{C}_{11}$	$\mathfrak{T}_2$	$\mathfrak{C}_{12}$	$\mathfrak{T}_1$	$\mathfrak{C}_{11}$	$\mathfrak{T}_1$	$\mathfrak{C}_{11}$
$\mathfrak{C}_{12}$	$\mathfrak{T}_1$		$\mathfrak{T}_3$		$\mathfrak{T}_3$	
$\mathfrak{C}_{13}$	$\mathfrak{T}_2$		$\mathfrak{T}_1$		$\mathfrak{T}_4$	
$\mathfrak{C}_{21}$	$\mathfrak{T}_3$	$\mathfrak{C}_{23}$	$\mathfrak{T}_1$	$\mathfrak{C}_{22}$	$\mathfrak{T}_1$	$\mathfrak{C}_{21}$
$\mathfrak{C}_{22}$	$\mathfrak{T}_3$		$\mathfrak{T}_1$		$\mathfrak{T}_2$	
$\mathfrak{C}_{23}$	$\mathfrak{T}_1$		$\mathfrak{T}_2$		$\mathfrak{T}_1$	
$\mathfrak{C}_{31}$	$\mathfrak{T}_1$	$\mathfrak{C}_{31}$	$\mathfrak{T}_2$	$\mathfrak{C}_{33}$	$\mathfrak{T}_1$	$\mathfrak{C}_{32}$
$\mathfrak{C}_{32}$	$\mathfrak{T}_4$		$\mathfrak{T}_1$		$\mathfrak{T}_1$	
$\mathfrak{C}_{33}$	$\mathfrak{T}_4$		$\mathfrak{T}_1$		$\mathfrak{T}_2$	
$\mathfrak{C}_{41}$	$\mathfrak{T}_1$	$\mathfrak{C}_{42}$	$\mathfrak{T}_3$	$\mathfrak{C}_{42}$	$\mathfrak{T}_1$	$\mathfrak{C}_{41}$
$\mathfrak{C}_{42}$	$\mathfrak{T}_1$		$\mathfrak{T}_1$		$\mathfrak{T}_2$	
$\mathfrak{C}_{51}$	$\mathfrak{T}_2$	$\mathfrak{C}_{53}$	$\mathfrak{T}_3$	$\mathfrak{C}_{53}$	$\mathfrak{T}_1$	$\mathfrak{C}_{52}$
$\mathfrak{C}_{52}$	$\mathfrak{T}_1$		$\mathfrak{T}_3$		$\mathfrak{T}_1$	
$\mathfrak{C}_{53}$	$\mathfrak{T}_1$		$\mathfrak{T}_1$		$\mathfrak{T}_3$	

The subcriteria's weights are established in a similar manner. Each expert determines the level of relevance for each subcriterion as in Table 5, and the final weights produced using the devised method are shown in Table 6 and figure 4

5.1.2 Evaluation of optimal energy source

The performance rating for each alternative against each criteria is provided by each expert from the Table 3 which results in the decision matrix as presented in Table 7. The individual inputs are coupled together using the BCLDFWG operators defined in eqn (2). The experts are treated with equal significance and the resulting score matrix determined based on eqn (3) is presented in Table 8. Following eqns (14) - (16) the normalized score matrix and then the associated normalized aggregated matrix for $\beta = 0.33$ is calculated. For determining the

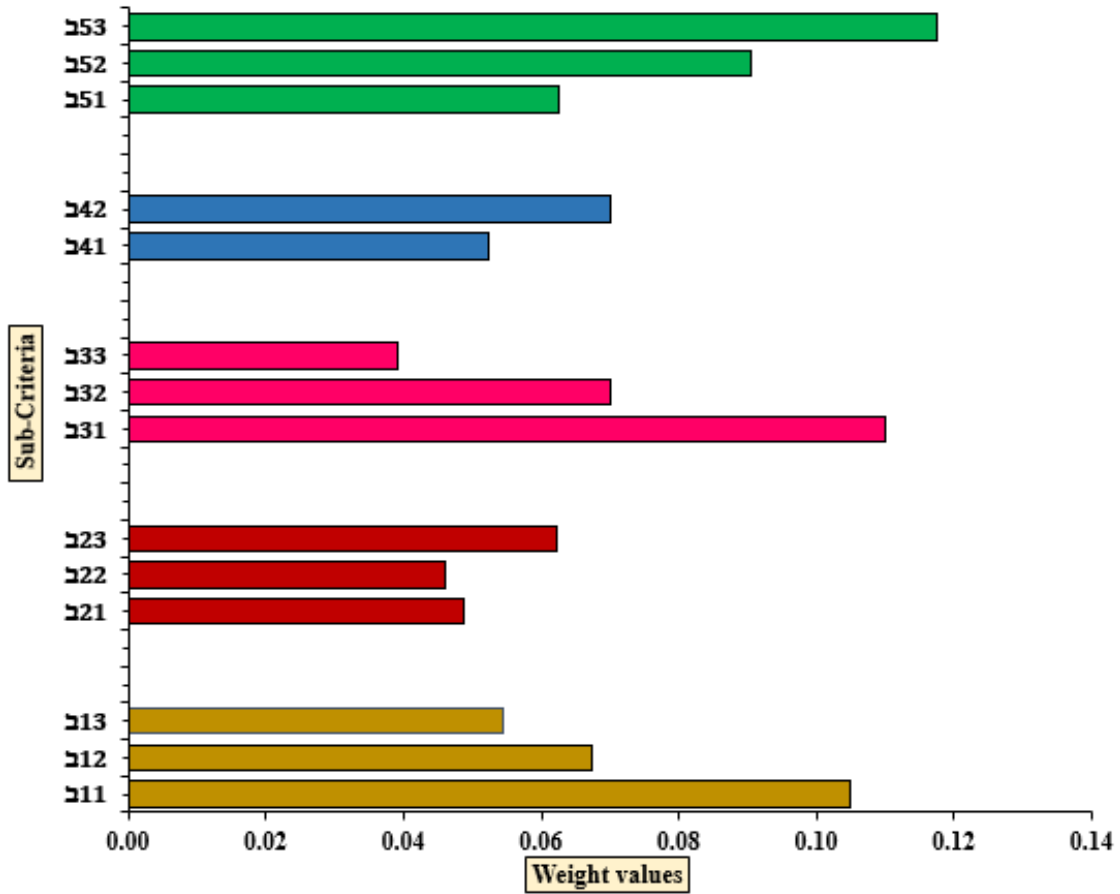


Figure 4: Weights of the attributes

519 entries of the satisfaction degree matrix, the constraint values for the attributes are defined
 520 by the experts as in Table 9. Using this Table 9 and the eqn (19)- (20), a constraint based
 521 normalization matrix is obtained.

522 Next, following the eqn (21) to (22), and incorporating the weight vectors from Table
 523 6, a constrained aggregated normalized decision matrix is determined. At the end, Step 7
 524 indicate a two stage distance assessment of the alternatives from the negative ideal. Table 10
 525 presents the normalized distance measures using eqn (23) - (29) and the prioritization values
 526 used in determining the most predominant energy source. The ranking of the alternatives
 527 are given in figure 5

Table 6: Weight vectors of the sub-criteria

Criteria	Sub-criteria	$\mathfrak{P}_1$	$\mathfrak{P}_2$	$\mathfrak{P}_3$	Local weights	Global weights (v_k)
$\mathfrak{C}_1$ (0.2268)	$\mathfrak{C}_{11}$	0.2377	0.4878	0.6632	0.4629	0.1050
	$\mathfrak{C}_{12}$	0.5545	0.1463	0.1895	0.296767	0.0673
	$\mathfrak{C}_{13}$	0.2079	0.3659	0.1474	0.2404	0.0545
$\mathfrak{C}_2$ (0.1571)	$\mathfrak{C}_{21}$	0.1734	0.2958	0.459	0.3094	0.0486
	$\mathfrak{C}_{22}$	0.1908	0.4930	0.1967	0.2935	0.0461
	$\mathfrak{C}_{23}$	0.6359	0.2113	0.3443	0.3972	0.0624
$\mathfrak{C}_3$ (0.2193)	$\mathfrak{C}_{31}$	0.6920	0.4706	0.3443	0.5023	0.1101
	$\mathfrak{C}_{32}$	0.1483	0.3530	0.4590	0.3201	0.0702
	$\mathfrak{C}_{33}$	0.1597	0.1765	0.1967	0.1776	0.0390
$\mathfrak{C}_4$ (0.1266)	$\mathfrak{C}_{41}$	0.5714	0.2222	0.4493	0.4143	0.0524
	$\mathfrak{C}_{42}$	0.4286	0.7778	0.4567	0.5544	0.0702
$\mathfrak{C}_5$ (0.2703)	$\mathfrak{C}_{51}$	0.1765	0.1734	0.3443	0.2314	0.0624
	$\mathfrak{C}_{52}$	0.3530	0.1908	0.4590	0.3343	0.0904
	$\mathfrak{C}_{53}$	0.4706	0.6359	0.1967	0.4344	0.1174

Table 7: Initial decision matrix based on each expert

Criteria/ Alternatives	$\mathfrak{C}_{11}$	$\mathfrak{C}_{12}$	$\mathfrak{C}_{13}$	$\mathfrak{C}_{21}$	$\mathfrak{C}_{22}$	$\mathfrak{C}_{23}$	$\mathfrak{C}_{31}$
$\mathfrak{C}_1$	(AI,VI,EI)	(VI,AI,MI)	(MI,VI,MU)	(MI,VU,MI)	(EI,EI,EI)	(MU,EI,MI)	(VI,MI,MU)
$\mathfrak{C}_2$	(VI,AI,AI)	(EI,VI,EI)	(VI,MI,MI)	(MI,EI,EI)	(E,MI,MI)	(EI,E,VI)	(AI,VI,MI)
$\mathfrak{C}_3$	(MI,MU,EI)	(MU,VU,MU)	(EI,EI,AU)	(MU,MU,VU)	(VU,AU,EI)	(AU,VU,MU)	(EI,MI,EI)
$\mathfrak{C}_4$	(VI,MI,VI)	(MI,EI,AI)	(MI,MI,EI)	(VU,EI,MU)	(AU,MU,VI)	(EI,MI,EI)	(MI,VI,VI)
$\mathfrak{C}_5$	(EI,EI,EI)	(MU,MI,VU)	(VU,MU,MU)	(AU,EI,VU)	(MU,VU,EI)	(VU,MU,MU)	(EI,MI,MI)

Criteria/ Alternatives	$\mathfrak{C}_{32}$	$\mathfrak{C}_{33}$	$\mathfrak{C}_{41}$	$\mathfrak{C}_{42}$	$\mathfrak{C}_{51}$	$\mathfrak{C}_{52}$	$\mathfrak{C}_{53}$
$\mathfrak{C}_1$	(EI,MI,VI)	(VI,VI,MU)	(MI,MI,VI)	(MI,VI,AI)	(MI,MU,EI)	(VI,AI,VI)	(VI,EI,AI)
$\mathfrak{C}_2$	(VI,EI,AI)	(AI,VI,VI)	(EI,VI,AI)	(VI,MI,AI)	(EI,MU,MU)	(VI,VI,AI)	(AI,MI,AI)
$\mathfrak{C}_3$	(MI,MI,MU)	(MU,VU,EI)	(VU,EI,MI)	(MI,MU,EI)	(VI,VI,VI)	(MI,EI,MU)	(MI,MU,EI)
$\mathfrak{C}_4$	(MU,EI,MI)	(EI,MI,VI)	(MI,MI,B)	(VI,VI,MI)	(EI,MI,VI)	(EI,EI,EI)	(VI,VI,VI)
$\mathfrak{C}_5$	(VU,MU,EI)	(MI,AU,EI)	(EI,EI,MU)	(EI,MI,MU)	(MU,EI,MI)	(MU,MU,EI)	(EI,EI,MI)

6 Result Validation

6.1 Sensitivity Analysis

In order to assess the purpose of attributes in the decision-making process and comprehend the efficacy of the technique, we have taken into account four scenarios by changing the

Table 8: Score matrix for evaluation

Criteria/ Alternatives	ζ_{11}	ζ_{12}	ζ_{13}	ζ_{21}	ζ_{22}	ζ_{23}	ζ_{31}
$\mathcal{C}_1$	0.4003	0.3483	0.1678	-0.0224	-0.0025	0.0038	0.1463
$\mathcal{C}_2$	0.6216	0.1637	0.3127	0.0751	0.0806	0.0945	0.4733
$\mathcal{C}_3$	0.0038	-0.3086	-0.2821	-0.3086	-0.4272	-0.4916	0.0751
$\mathcal{C}_4$	0.3898	0.3272	0.1522	-0.2405	-0.1782	0.0751	0.3898
$\mathcal{C}_5$	-0.0025	-0.1633	-0.3086	-0.4272	-0.2405	-0.3086	0.1522

Criteria/ Alternatives	ζ_{32}	ζ_{33}	ζ_{41}	ζ_{42}	ζ_{51}	ζ_{52}	ζ_{53}
$\mathcal{C}_1$	0.2379	0.2465	0.3127	0.4733	0.0038	0.5430	0.4003
$\mathcal{C}_2$	0.4003	0.543	0.4003	0.4733	-0.1444	0.543	0.5539
$\mathcal{C}_3$	0.0806	-0.2405	-0.0969	0.0038	0.4626	0.0038	0.0038
$\mathcal{C}_4$	0.0038	0.2379	0.3127	0.3898	0.2379	-0.0025	0.4626
$\mathcal{C}_5$	-0.2405	-0.2100	-0.074	0.0038	0.0038	-0.1444	0.0751

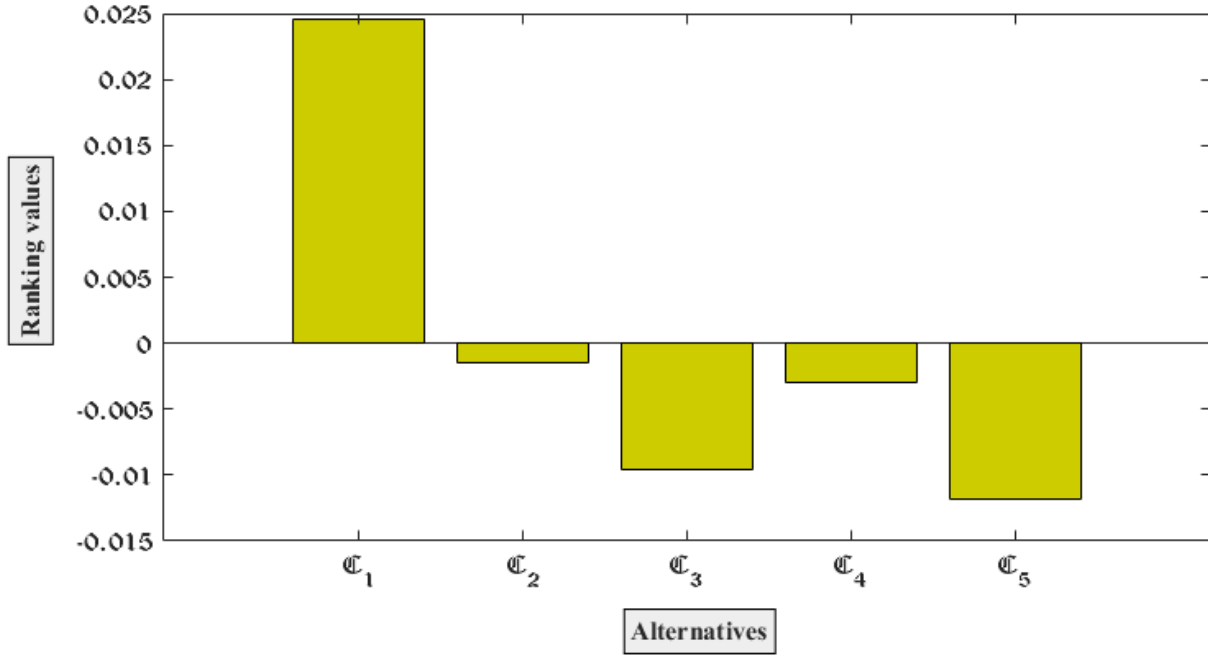


Figure 5: Combined score of alternatives

Table 9: Constraint value of the criteria

	ζ_{11}	ζ_{12}	ζ_{13}	ζ_{21}	ζ_{22}	ζ_{23}	ζ_{31}	ζ_{32}	ζ_{33}	ζ_{41}	ζ_{42}	ζ_{51}	ζ_{52}	ζ_{53}
LB_k	0.00	-0.25	Min	Min	-0.3	-0.49	Min	-0.2	Min	Min	0.1	Min	-0.1	0.2
UB_k	Max	Max	Max	Max	Max	Max	Max	Max	Max	Max	Max	Max	Max	Max

significance of the sub-criteria in the order of positive and negative criteria. The obtained findings are utilized to investigate how the weighted criteria affect the problem.

Case 1: 50% significance to positive and 50% to negative criteria i.e, for $\zeta_k \in \mathfrak{h}$,

Table 10: Results using improved TRUST approach

Alternatives	$\mathfrak{D}_i$	$\mathfrak{M}_i$	$\mathfrak{Z}_i$	$\mathfrak{N}_i$	$\sum_{r=1}^l \mathfrak{S}_{ir}$	$\sum_{r=1}^l \mathfrak{Q}_{ir}$	λ_i	Prioritization
$\mathfrak{C}_1$	0.0982	0.1394	0.0372	0.1302	0.0188	0.0303	0.0246	1
$\mathfrak{C}_2$	0.0916	0.1256	0.0338	0.0293	-0.0146	0.0116	-0.0015	2
$\mathfrak{C}_3$	0.0961	0.1098	0.026	0.1304	0.0079	-0.0271	-0.0096	4
$\mathfrak{C}_4$	0.0904	0.1254	0.0342	0.1121	-0.0206	0.0147	-0.003	3
$\mathfrak{C}_5$	0.0962	0.1074	0.025	0.1214	0.0084	-0.032	-0.0118	5

$v_k = 0.0455$ and the rest takes the value of 0.1667.

Case 2: 60% significance to positive (i.e, $v_k = 0.0546$) and 40% to negative criteria ($v_k = 0.0546$)

Case 3: 70% significance to positive (i.e, $v_k = 0.0636$) and 30% to negative criteria ($v_k = 0.1$)

Case 4: 80% significance to positive (i.e, $v_k = 0.0727$) and 20% to negative criteria ($v_k = 0.0667$)

Table 11: Sensitive ranking of alternatives

Case 1		Case 2		Case 3		Case 4	
Ranking values of $\mathfrak{C}_i$	Preference	Ranking values of $\mathfrak{C}_i$	Preference	Ranking values of $\mathfrak{C}_i$	Preference	Ranking values of $\mathfrak{C}_i$	Preference
0.2659	1	0.2137	1	0.1633	1	0.1132	1
-0.0728	5	-0.0576	5	-0.041	5	-0.0222	3
-0.0358	3	-0.0314	2	-0.0302	2	-0.0283	4
-0.0577	4	-0.046	4	-0.0337	4	-0.0204	2
-0.0328	2	-0.0325	3	-0.0322	3	-0.031	5

As shown in the table 11 and figure 6, the criteria weights have a considerable impact on the favoured alternative. However, the most chosen choice from the four scenarios aligns with the recommended ranking, with a significant impact on the rest of the favoured options. As a result, the obtained results illustrate the susceptible nature of the proposed methodology in selecting the optimal option.

6.2 Comparison Analysis

To assess the trustworthiness of the created framework, the obtained results are compared to various distance-based ranking methodologies such as VIKOR, TOPSIS, CODAS, EDAS, and the traditional TRUST methods. The table 12 and figure 7 shows that the outcomes do not correspond with the preferred option, but have resulted in the attainment of the nearest possible option. Unlike the previous MADM approaches mentioned, the use of three

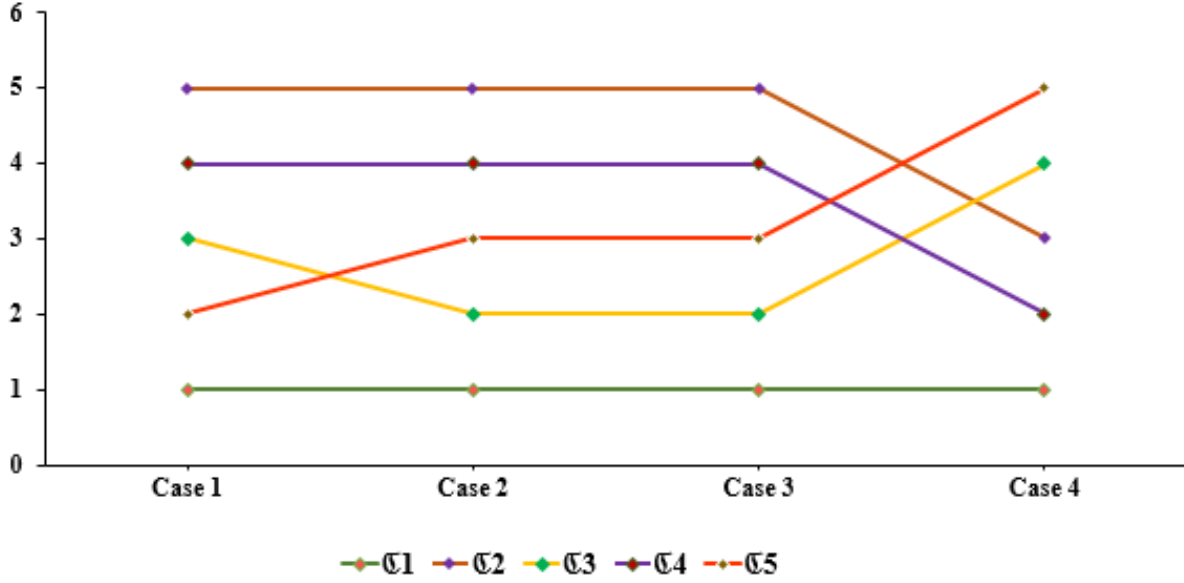
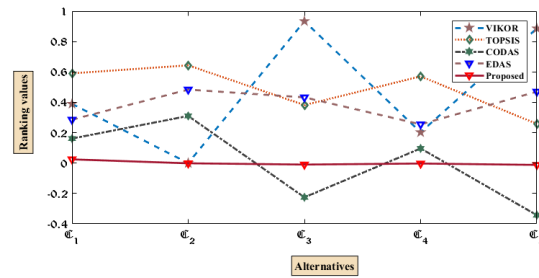
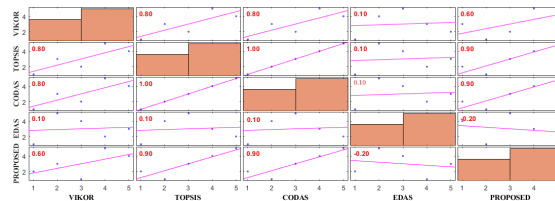


Figure 6: Ranking order of alternatives

stage normalisation of the original information and aggregation of distance measurements in ranking the alternatives shows the suggested framework's distinctiveness in methodically addressing the scenario at hand. Furthermore, when dealing with negative information, the classical TRUST technique loses its versatility, and the presence of a large number of choices and qualities restricts the method's efficacy.



(a) Outcomes from various MADM techniques



(b) Spearman rank correlation matrix

Figure 7: Comparison ranking results

Table 12: Comparative ranking of alternatives

MADM approaches	Ranking values	Preferred Option
VIKOR	0.3897, 0, 0.9338, 0.2027, 0.8869	$\mathfrak{C}_1$
TOPSIS	0.5906, 0.6434, 0.3819, 0.5706, 0.2583	$\mathfrak{C}_1$
CODAS	0.1615, 0.3102, -0.2253, 0.0964, -0.3428	$\mathfrak{C}_1$
EDAS	0.2858, 0.4837, 0.4325, 0.2551, 0.4702	$\mathfrak{C}_1$
Classical TRUST	Cannot be determined	-

To further investigate the relationship between the examined MADM approaches and the proposed method, the spearman rank correlation is used, and the results are shown in the figure. The TOPSIS and CODAS approaches have a notably greater association, the VIKOR approach has a moderate relationship, and the EDAS method has a weak relationship with the suggested study. Thus, the combination of the LBWA technique and the use of BCLDF logic in describing imprecise information has impacted the dependability of the systematic frameworks.

6.3 Discussion

The major factors that were looked at for evaluation using the LBWA technique are organised in the following order of importance: environmental, technological, social, economic, and political challenges. The table 6 also shows that the power plant's output capacity is the most important aspect in the context of technical factors, followed by the system's effectiveness and maturity. Similar to this, it is thought that the price per kg of hydrogen produced, which depends on operations and maintenance, is significantly influenced by budgetary constraints. Since sophisticated technology should be able to offer employment chances that will promote the country's economic well-being, job creation is regarded as being the most significant of the social variables taken into account. Additionally, the sub-criteria most pliable to governmental policy are those in the condition of political limits. Likewise the environmental criteria place a strong emphasis on the sustainability of an energy source, which includes the resource's long-term availability, followed by pollutant emission and land use as the most important considerations.

According to the evaluation results produced using the improved fuzzy TRUST technique, solar energy has taken a significantly higher priority over other sources of hydrogen

581 production in India. India will be a major player in the transition to a hydrogen economy
582 because it has the most solar PVs deployed. Further it also offers many advantages, in-
583 cluding its wide availability, technological advancements, government initiatives that makes
584 it affordable, and its minimal impact on the environment. Furthermore, renewable energy
585 sources such as wind and hydro are favoured, whilst biomass and waste-to-energy sources
586 are ranked as the least preferred. Wind energy has similar benefits to solar energy, and the
587 government-led "Make in India" project has helped the country create wind farms and tur-
588 bines, making wind-based H₂ generation economical. Hydropower-based energy production
589 is another well-established system that considerably adds to the nation's power output. In
590 comparison to other energy sources, biomass and WtE systems are blooming technologies
591 with limitations due to technological, economic, and environmental challenges. Nonethe-
592 less, attempts are being made to exploit the potential of these renewable energy sources in
593 producing H₂ via cost-effective, high-efficiency, low-emission systems.

594 It is clear from the findings that solar, wind, and hydro energy represent the best
595 prospects for producing H₂ in India. The study's findings can help professionals and policy-
596 makers choose a suitable RES for creating green H₂ in the country.

597 7 Conclusion

598 The Indian energy sector is undertaking a transition towards cleaner development, with
599 the goal of achieving energy independence by 2070. Building a hydrogen economy based
600 on RES will have a substantial impact not only on environmental sustainability, but also
601 on economic growth, foreign investment, the standard of living, policy stability, and job
602 opportunities in the country. The suggested study created a detailed evaluation of the RES
603 for hydrogen production in India. An integrated BCLDF framework based on an expert
604 viewpoint is used to evaluate clean and sustainable sources. The LBWA approach is used
605 to find the relevant criteria for problem evaluation based on the experts' choices and past
606 studies. The improved TRUST technique is also used to determine which of the five energy
607 sources is the most environmentally benign and economically viable. The inputs for the
608 alternative are provided using BCLDF logic, which provided more space in expressing the
609 fuzziness and two-sided information of the alternatives from the decision-maker's perspective,
610 which played a significant role in achieving an acceptable solution. Solar energy, followed by
611 wind and hydropower, is the best choice for H₂ production in India. The sensitive analysis

demonstrates the importance of the attributes in the choice process, and the comparative study demonstrates the superiority of the created method in achieving trustworthy solutions when compared to other options distance-based assessment methodologies. The method is also superior to others due to the multistage normalisation of the input data, which has aided in achieving a consistent result, and the two-stage distance evaluation helped in attaining robust solution.

Furthermore, the study assists policymakers and stakeholders in determining the most efficient energy source for hydrogen generation in the country. It can also aid in attracting international investments and prioritising policy implications, which will foster the growth of the hydrogen economy and so contribute to the nation's growth. Furthermore, the cost of electrolyzers utilised in the electrolytic process for generation is a crucial aspect that requires additional exploration for the sector's development. Nevertheless, the produced study's versatility is slightly limited by the study's entire reliance on the expert opinion, which eliminates the objectivity of the analysis.

The following scopes are recommended. The versatility of the developed BCLDFS in coping with complicated circumstances will be investigated. Given the benefits of fuzzy-based frameworks in dealing with multifaceted circumstances, integrating them into the analysis of hydrogen transit and storage barriers would be beneficial. Further, extending the problem to examine hydrogen production using hybrid systems, as well as the barriers to deployment through a decision analysis, can assist the sector in achieving long-term growth.

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Declaration

Availability of Data and Materials

All data generated or analyzed during this study are included in this article.

Competing interests

The authors declare that they have no competing interests.

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Authors' Contribution

Joseph Raj Vikilal Joice Brainy: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing - original draft.

Samayan Narayanamoorthy: Writing - original draft, Data curation, Methodology, Software, Validation, Writing - review and editing.

Subramaniam Pragathi: Writing - original draft, Data curation, Methodology, Software, Validation, Writing - review and editing.

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Availability of data

The data that supports the findings of this study are available within the article.

Ethics declarations

Ethics approval and consent to participate

Not applicable

Consent for publication

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Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Highlights:

- To analyse the potential of renewable sources in hydrogen production
- A novel concept of BCLDFS for handling uncertainty is developed
- An enhanced fuzzy LBWA-TRUST model is developed
- To validate the fuzzy model, green hydrogen opportunities for India is examined